SPmodels: Technical Mathematical Derivations

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Matérn correlation: derivation and implementation

Question, scope and prerequisites

This chapter derives the unit-variance Matérn correlation implemented by `matern_cor()`. It answers: given two point-supported observations separated by a non-negative distance, how strongly are their latent spatial effects correlated under a stationary, isotropic Matérn model?

The chapter assumes introductory probability, calculus and linear algebra. Readers need to know that a covariance matrix must be symmetric and positive semidefinite. The modified Bessel function of the second kind, written `K_nu(x)`, is introduced where it is used.

This first slice implements correlation evaluation only. It does not yet implement a variance multiplier, nugget or measurement-error term, covariance derivatives, model fitting, prediction, or a practical-range conversion.

Notation and assumptions

Let `s_i` and `s_j` be locations in a Euclidean coordinate system whose distance unit is known. Define

- `d_ij = ||s_i - s_j|| >= 0`, the distance between the locations;
- `a > 0`, the Matérn scale in the same units as `d_ij`;
- `nu > 0`, the dimensionless smoothness parameter; and
- `x_ij = d_ij / a`, a dimensionless distance.

We assume stationarity, so covariance does not depend on absolute location, and isotropy, so it depends on separation only through distance. Coordinates in angular degrees must not be passed as though they were planar distances.

Correlation definition

For positive distance, SPmodels uses the scale parameterisation

(M-01)

```
rho(d; a, nu) = [2^(1-nu) / Gamma(nu)] x^nu K_nu(x),
```

where `x = d/a`, `Gamma` is the gamma function, and `K_nu` is the modified Bessel function of the second kind. Calling `a` a scale rather than a range is deliberate: different fields use incompatible range conventions.

Value at zero

For `nu > 0`, the small-argument behaviour of the Bessel function is

`K_nu(x) ~ 2^(nu-1) Gamma(nu) x^(-nu)` as `x` tends to zero from above.

Substitution into (M-01) gives

'rho(d; a, nu) [2^(1-nu) / Gamma(nu)] x^nu [2^(nu-1) Gamma(nu) x^(-nu)] = 1'.

Therefore the continuous definition is

(M-02) `rho(0; a, nu) = 1`.

The implementation assigns this value directly. It does not evaluate a singular Bessel expression at zero.

Units and rescaling

Because distance and scale enter only through their ratio,

(M-03) `rho(c d; c a, nu) = rho(d; a, nu)` for every `c > 0`.

Thus converting both metres to kilometres leaves the correlation unchanged. Converting only one of them changes the model and is an input error that a future geometry layer must prevent.

Half-integer checks

Half-integer Bessel functions have elementary forms. For `nu = 1/2`,

`K_(1/2)(x) = sqrt(pi / (2x)) exp(-x)` and `Gamma(1/2) = sqrt(pi)`.

Putting these identities into (M-01),

`rho = [2^(1-1/2)/sqrt(pi)] x^(1/2)  sqrt(pi/(2x)) exp(-x)  = exp(-x)`.

Hence

(M-04) `rho(d; a, 1/2) = exp(-d/a)`.

The recurrence relations for half-integer Bessel functions similarly give

(M-05) `rho(d; a, 3/2) = (1 + x) exp(-x)`, and

(M-06) `rho(d; a, 5/2) = (1 + x + x^2/3) exp(-x)`.

These three identities provide calculations independent of the general Bessel formula and are used in deterministic tests.

From a correlation to a matrix

For locations `s_1, ..., s_n`, define `R_ij = rho(d_ij; a, nu)`. Euclidean distance is symmetric, so `d_ij = d_ji` and consequently `R_ij = R_ji`. Equation (M-02) gives `R_ii = 1`.

The Matérn family has a non-negative spectral density; by the spectral characterisation of stationary covariance functions, its inverse Fourier transform is positive definite. Therefore, for every real vector `z`,

(M-07) `z' R z >= 0`.

Finite-precision eigenvalues can be slightly negative, so tests allow only a small predeclared numerical tolerance. This mathematical property applies when the supplied distances come from a valid Euclidean configuration; an arbitrary non-negative matrix need not be a valid distance matrix.

Numerical algorithm

Directly evaluating (M-01) can overflow near zero because `K_nu(x)` diverges, and can underflow at large `x` because it decays exponentially. R provides the exponentially scaled value

(M-08) `K_scaled(x, nu) = exp(x) K_nu(x)`.

For positive `x`, take logarithms of (M-01) and use (M-08):

(M-09)

`log rho = (1-nu) log 2 - log Gamma(nu) + nu log x + log K_scaled(x, nu) - x`.

The implementation evaluates `log Gamma(nu)` with `lgamma()`, evaluates the scaled Bessel function with `besselK(..., expon.scaled = TRUE)`, and exponentiates only at the end. Underflow to zero at very large distance is a valid floating-point representation of a correlation smaller than the available precision. At extremely small positive arguments, a Bessel overflow is replaced by the limiting value one only below machine epsilon. Any other non-finite result is an explicit error. Values displaced a few floating-point units beyond the theoretical interval are clipped to `[0, 1]`.

For the exact half-integer values `nu = 1/2, 3/2, 5/2`, implementation uses (M-04)–(M-06) directly. This is algebraically identical to the generic definition and avoids repeated Bessel evaluation during covariance optimization. Every other positive smoothness retains the scaled-Bessel log path.

Algorithm:

1. Validate that all distances are finite and non-negative.
2. Validate that `a` and `nu` are finite positive scalars.
3. Form `x = d/a`.
4. Assign one where `x = 0` using (M-02).
5. For positive `x`, use (M-04)–(M-06) at their exact half-integer smoothness values; otherwise evaluate (M-09).
6. Apply the documented extreme-small-argument recovery and mathematical bounds, preserving the input dimensions and names.

The algorithm is deterministic and uses no random numbers.

Interpretation and limitations

Larger scale makes a fixed physical distance smaller on the dimensionless scale and therefore usually increases correlation. Smoothness controls both short-distance behaviour and differentiability of the

associated Gaussian field. It is not a variance parameter. A complete covariance model will later combine this correlation with separately named spatial variance, nugget and measurement-error components.

This implementation assumes distances have already been computed correctly. It does not inspect coordinate reference systems, convert units, validate that a supplied matrix is Euclidean, or determine whether points and polygons have compatible support.

Implementation and validation map

Equations	Implementation	Test evidence
M-01, M-02, M-08, M-09	`R/matern.R`, `matern_cor()`	zero value and general evaluation
M-03	same	joint distance/scale rescaling
M-04–M-06	same	three half-integer closed forms
M-07	same, applied elementwise to a distance matrix	symmetry, unit diagonal and eigenvalue bound

The tests also cover dimension/name preservation and invalid distances, scales and smoothness values. Reference-implementation comparison remains planned in the separate pinned validation environment; no agreement claim is made here.

The public `matern_cor()` function performs the complete contract validation above. During model fitting, distances and scalar covariance parameters have already passed those checks before numerical optimization begins. The internal covariance builder therefore uses the identical closed forms for $\nu = 1/2, 3/2, 5/2$, and the public generic Bessel calculation otherwise, without repeatedly scanning the whole distance matrix for the same input conditions. This changes validation placement rather than the mathematical function. Tests compare the internal and public results element by element for specialized and generic smoothness values and require preservation of matrix dimensions and labels.

Worked calculation

Take two locations 500 metres apart, scale `a = 1000` metres and `nu = 1/2`. Then `x = 500/1000 = 0.5`. By (M-04),

`rho = exp(-0.5) = 0.6065307` (rounded).

Changing both quantities to kilometres gives `d = 0.5`, `a = 1` and the same answer, illustrating (M-03).

Exercises

1. Use (M-04) to find the distance at which correlation is `0.05` when `a = 2 km`. Answer: solve `exp(-d/a) = 0.05`, giving `d = -2 log(0.05) km`.
2. Verify algebraically that (M-05) equals one at zero.
3. Explain why applying (M-01) to longitude differences measured in degrees and a scale measured in kilometres violates (M-03)'s unit requirement.
4. Construct three one-dimensional locations, calculate their distance matrix, and verify numerically that the resulting correlation matrix has no materially negative eigenvalue.

References

- M. L. Stein (1999), *Interpolation of Spatial Data: Some Theory for Kriging*, Springer, sections on the Matérn class.
- C. E. Rasmussen and C. K. I. Williams (2006), *Gaussian Processes for Machine Learning*, MIT Press, covariance functions chapter.
- NIST Digital Library of Mathematical Functions, chapters 5 and 10, gamma and modified Bessel functions: https://dlmf.nist.gov/.

Gaussian geostatistical regression: derivation and algorithm

Purpose and prerequisites

This chapter derives the first complete SPmodels fitting path. It assumes introductory linear models, multivariate normal distributions, matrix multiplication and logarithms. The implementation is independent and follows the equations below directly.

The current path answers three questions:

1. How is a linear mean combined with spatially correlated residual variation?
2. How are regression and covariance parameters fitted by maximum likelihood?
3. How are predictions and their conditional uncertainties computed?

It does not yet cover non-Gaussian responses, anisotropy, spatio-temporal dependence, repeated observations, restricted maximum likelihood or covariance parameter uncertainty.

Data, assumptions and notation

For locations $s_1, \dots, s_n$, let Y_i be a continuous response and let the row vector x_i^T contain its mean-model predictors. Stack these into the response vector y and the full-rank $n \times p$ design matrix X. An optional known offset is the vector o.

The model is

$$Y = X\beta + o + S + \epsilon, \tag{G-01}$$

where

$$S \sim N(0, \sigma_s^2 R), \qquad \epsilon \sim N(0, \tau^2 I), \qquad S \perp \epsilon. \tag{G-02}$$

The assumptions are:

- the conditional response is Gaussian with identity link;
- both supplied coordinate axes are Euclidean and use the explicitly recorded common unit;
- dependence is stationary and isotropic, so it depends only on distance;
- the Matérn smoothness $\nu > 0$ is fixed;
- $\sigma_s^2 > 0$, $\rho > 0$, and $\tau^2 > 0$;
- the observation-level errors are mutually independent and independent of the spatial process;
- the design matrix has full column rank.

Here σ_s^2 is the spatial variance, ρ is the Matérn scale and τ^2 is the nugget variance. The distinction matters: the latent process has variance σ_s^2, while an observation has marginal variance $\sigma_s^2 + \tau^2$.

From coordinates to a covariance matrix

For two coordinate vectors $s_i = (s_{i1}, s_{i2})^T$ and $s_j = (s_{j1}, s_{j2})^T$, the Euclidean distance is

$$d_{ij} = \sqrt{(s_{i1} - s_{j1})^2 + (s_{i2} - s_{j2})^2}. \tag{G-03}$$

Let $M_\nu(d/\rho)$ be the unit-variance Matérn correlation derived in `01_MATERN_CORRELATION.md`. Then

$$R_{ij} = M_\nu(d_{ij}/\rho), \tag{G-04}$$

and, after adding the independent nugget,

$$V = \mathrm{Var}(Y) = \sigma_s^2 R + \tau^2 I. \tag{G-05}$$

Because the nugget is strictly positive, V is positive definite in exact arithmetic even when locations are duplicated. The initial implementation still requires at least two distinct locations because a spatial scale cannot be learned or meaningfully checked when every distance is zero.

Gaussian log-likelihood

Define the offset-adjusted response $y_o = y - o$. From (G-01)–(G-05),

$$Y \sim N(X\beta + o, V). \tag{G-06}$$

The multivariate normal density gives

$$\ell(\beta, \theta) = -\frac{1}{2}\left\{n\log(2\pi) + \log|V| + (y_o - X\beta)^T V^{-1}(y_o - X\beta)\right\}, \tag{G-07}$$

where $\theta = (\sigma_s^2, \rho, \tau^2)^T$, with ν fixed.

Independent limiting model

With `spatial = FALSE`, the spatial process is absent and

$$Y = X\beta + o + \varepsilon, \qquad \varepsilon \sim N(0, \tau^2 I). \tag{G-06a}$$

Equations G-07–G-09 then reduce to ordinary least squares, while maximum likelihood estimates

$$\hat{\tau}^2_{\mathrm{ML}} = \frac{1}{n}(y_o - X\hat{\beta})^T(y_o - X\hat{\beta}). \tag{G-06b}$$

The divisor is n, not $n - p$, because this is the maximum-likelihood variance estimator. This route supplies an independent limiting check and a non-spatial comparison without pretending that a spatial fit is an ordinary linear model.

For any fixed θ, differentiate (G-07) with respect to β:

$$\frac{\partial \ell}{\partial \beta} = X^T V^{-1}(y_o - X\beta). \tag{G-08}$$

Setting (G-08) to zero produces the generalized least-squares estimator

$$\hat{\beta}(\theta) = (X^T V^{-1} X)^{-1} X^T V^{-1} y_o. \tag{G-09}$$

Substituting (G-09) into (G-07) produces the profile log-likelihood $\ell_p(\theta) = \ell\{\hat{\beta}(\theta), \theta\}$. SPmodels maximises this ordinary maximum likelihood criterion over the logarithms of the three positive covariance parameters. Log parameters enforce positivity because $\exp(a) > 0$ for every finite a. This is ML, not REML.

The default starting scale is the median positive pairwise distance. An ordinary least-squares residual variance supplies starting variance values, split 80% spatial and 20% nugget. User-supplied positive starting values take precedence. Bounds are deliberately broad safeguards against numerical overflow, not inferential constraints.

Stable computation without an explicit inverse

Writing V^{-1} makes the algebra readable, but explicitly computing an inverse is unnecessary and can amplify rounding error. Since V is positive definite, its Cholesky factorisation is

$$V = LL^T \tag{G-10}$$

(or equivalently $V = U^T U$ for the upper-triangular convention used by R). To obtain $V^{-1}b$, solve two triangular systems. Also,

$$\log |V| = 2 \sum_{i=1}^{n} \log L_{ii}. \tag{G-11}$$

The same approach is used for $X^T V^{-1} X$. A failed Cholesky factorisation means the proposed numerical system is not usable; the optimiser rejects that proposal. The implementation does not silently add jitter because doing so would change the stated statistical model. Rejection uses the finite objective sentinel 10^{20}, avoiding overflow in bounded numerical differentiation near an invalid trial point. The final parameter vector must still produce valid Cholesky factors.

Conditional on fitted covariance parameters,

$$\mathrm{Var}(\hat{\beta} \mid \hat{\theta}) = (X^T V^{-1} X)^{-1}. \tag{G-12}$$

The `vcov()` and coefficient standard errors report (G-12). They do not propagate uncertainty in $\hat{\theta}$.

Approximate covariance-parameter uncertainty

The optimiser works with $\eta = (\log \sigma_s^2, \log \rho, \log \tau^2)^T$. Let H_η be the numerical Hessian of the negative profile log-likelihood at the optimum. When it is finite and positive definite, the observed-information approximation is

$$\mathrm{Var}(\hat{\eta}) \approx H_\eta^{-1}. \tag{G-12a}$$

For $\theta = \exp(\eta)$, the delta-method Jacobian is $J = \mathrm{diag}(\hat{\theta})$. Therefore

$$\boxed{\mathrm{Var}(\hat{\theta}) \approx J H_\eta^{-1} J^T.} \tag{G-12b}$$

The square roots of the diagonal of (G-12b) appear in `summary()`. If H_η is not positive definite, they are reported as missing rather than created by silently altering the Hessian. These are local large-sample approximations and do not include boundary corrections.

Conditional spatial prediction

Consider a new location s_*, predictor row x_*^T, and offset o_*. Let

$$c_* = \mathrm{Cov}(Y, S_*) = \sigma_s^2\{M_\nu(d(s_1, s_*)/\rho), \ldots, M_\nu(d(s_n, s_*)/\rho)\}^T. \tag{G-13}$$

The nugget is absent from (G-13): a future independent observation error is not shared with the observed errors. The Gaussian conditioning identity gives the universal-kriging predictor

$$\widehat{Y_*} = x_*^T\hat{\beta} + o_* + c_*^T V^{-1}(y_o - X\hat{\beta}). \tag{G-14}$$

For known β, conditioning reduces the latent variance from σ_s^2 by $c_*^T V^{-1} c_*$. Estimating β adds the universal-kriging correction. Define

$$q_* = x_* - X^T V^{-1} c_*. \tag{G-15}$$

Then the reported latent prediction variance is

$$v_{*,\mathrm{latent}} = \sigma_s^2 - c_*^T V^{-1} c_* + q_*^T (X^T V^{-1} X)^{-1} q_*. \tag{G-16}$$

For a future noisy observation, its independent nugget is added:

$$v_{*,\mathrm{observation}} = v_{*,\mathrm{latent}} + \tau^2. \tag{G-17}$$

`predict(..., se.fit = TRUE, target = "latent")` uses (G-16), while `target = "observation"` uses (G-17). Both condition on fitted covariance parameters. Therefore these standard errors are not complete parameter- uncertainty intervals and must not be described as such.

Algorithm implemented

1. Construct the model frame using ordinary R formula rules, apply the subset and missing-value policy, and build X.
2. Match those rows to the two numeric coordinate columns and verify finite values, explicit units, full-rank X, and at least two distinct sites.
3. Compute the Euclidean distance matrix once using (G-03).
4. For fixed covariance values, proceed directly to step 6. Otherwise construct starting values and optimise the negative profile log-likelihood on the log parameter scale.
5. At every objective evaluation, build (G-05), factor it, calculate (G-09), and evaluate (G-07) using (G-10)–(G-11).
6. Store the final triangular factors and matrix products needed by standard methods.
7. For prediction, build cross-distances and (G-13), then evaluate (G-14)–(G-17) by triangular solves.

The direct dense factorisation requires approximately $O(n^3)$ computation and $O(n^2)$ memory. Large-data approximations are future capabilities and must be derived and validated separately.

Independent checks and interpretation

`tests/test-gaussian-fit.R` checks this implementation against matrix equations written directly with base R:

- generalized least squares (G-09);
- log-likelihood (G-07);
- conditional mean (G-14);
- latent and observation variances (G-16)–(G-17);
- invariance to row permutation;
- positivity and convergence of an estimated covariance fit;
- explicit failures for unsupported or ambiguous input.

The half-integer identity $M_{1/2}(d/\rho) = \exp(-d/\rho)$ makes the test covariance independently transparent. These are local mathematical checks, not evidence of agreement with the separately managed reference implementation.

Code map

Equations or task	Implementation
G-03	`.sp_dist()` in `R/fit.R`
G-04–G-05	`.sp_covariance()` in `R/fit.R` and `matern_cor()`
G-07–G-12	`.sp_gaussian_state()` and `sp_fit()` in `R/fit.R`
G-13–G-17	`predict.spmodels_fit()` in `R/methods.R`
Input and optimiser controls	`sp_control()` in `R/control.R`
Direct checks	`tests/test-gaussian-fit.R`

Practical cautions

- Coordinates such as longitude and latitude in degrees are not automatically projected. Supply appropriate Euclidean coordinates and state their units.
- An estimated scale is expressed in the supplied coordinate unit and is the parameter in $M_\nu(d/\rho)$, not a named practical range.
- A small dataset may weakly identify spatial variance, scale and nugget.
- A convergence code of zero is numerical evidence that the optimiser met its stopping rule, not proof of global optimality or a suitable scientific model.
- Residual diagnostics, covariance-parameter intervals and reference numerical comparisons remain required before research conclusions rely on this path.

Gaussian simulation, empirical variograms and residual diagnostics

Purpose

This chapter derives the three assessment operations added to the Gaussian geostatistical model: unconditional response simulation, an empirical residual variogram and decorrelated residuals. It also states what each operation can and cannot establish.

The notation follows the Gaussian model chapter:

$$Y = X\beta + o + S + \varepsilon, \qquad Y \sim N(\mu, V),$$

where

$$\mu = X\beta + o, \qquad V = \sigma_s^2 R + \tau^2 I.$$

Unconditional simulation

Let $V = LL^{\mathsf{T}}$ be a Cholesky factorisation and draw $z \sim N(0, I)$. Define

$$\boxed{Y^{(b)} = \hat{\mu} + \widehat{L} z^{(b)}.} \tag{A-01}$$

Because $E(z) = 0$,

$$E\{Y^{(b)}\} = \hat{\mu}. \tag{A-02}$$

Using $\mathrm{Var}(Az) = A\,\mathrm{Var}(z)A^{\mathsf{T}}$,

$$\mathrm{Var}\{Y^{(b)}\} = \widehat{L} I \widehat{L}^{\mathsf{T}} = \widehat{V}. \tag{A-03}$$

Thus (A-01) has exactly the fitted model's response mean and covariance. Each simulation column uses an independent standard-normal vector. The current `simulate()` method generates outcomes at the observed sites only. It is unconditional on the observed responses and conditional on fitted parameter values; it does not propagate parameter uncertainty.

When a seed is supplied, SPmodels temporarily sets it and restores the caller's previous random-number state. Without a supplied seed, ordinary R random-number state advances normally.

Empirical semivariance

For residuals e_i and e_j at sites separated by distance d_{ij}, the pairwise empirical semivariance is

$$\boxed{g_{ij} = \frac{1}{2}(e_i - e_j)^2.} \tag{A-04}$$

Split the interval from zero to a chosen maximum distance into K equal-width bins. Let P_k be the set of unordered pairs in bin k, and let $n_k = |P_k|$. The binned empirical variogram is

$$\boxed{\hat{\gamma}_k = \frac{1}{n_k} \sum_{(i,j) \in P_k} g_{ij}.} \tag{A-05}$$

SPmodels always returns n_k. A semivariance based on few pairs is unstable, so the pair count is part of the result rather than merely plotting metadata. Empty bins have a missing empirical value.

The default residual is the mean-model residual

$$e_{\text{mean}} = y - o - X\hat{\beta}. \tag{A-06}$$

This retains the spatial signal that the variogram is intended to reveal. The optional conditional residual subtracts the fitted conditional spatial effect as well. Its variogram answers a different diagnostic question: how much dependence remains after spatial prediction?

Fitted model variogram

For two distinct response observations under the stationary model,

$$\mathrm{Var}(Y_i - Y_j) = 2\{\sigma_s^2 + \tau^2 - \sigma_s^2 R(d_{ij})\}.$$

The theoretical semivariogram is half this quantity:

$$\boxed{\gamma(d) = \tau^2 + \sigma_s^2\{1 - R(d)\}, \qquad d > 0.} \tag{A-07}$$

At exactly zero lag the conventional value is $\gamma(0) = 0$, with a jump of size τ^2 immediately to the right when a nugget is present. SPmodels evaluates (A-07) at positive bin midpoints and places it beside (A-05). This is a descriptive diagnostic comparison, not a second fitting criterion.

Decorrelated residuals

Under the fitted Gaussian model, the mean-model residual vector should have approximately covariance V. If $V = LL^{\mathsf{T}}$, define

$$\boxed{z_{\mathrm{res}} = L^{-1}(y - o - X\widehat{\beta}).} \tag{A-08}$$

If the mean and covariance parameters were known and the model were correct, then

$$\mathrm{Var}(z_{\mathrm{res}}) = L^{-1} V L^{-\mathsf{T}} = I. \tag{A-09}$$

The components would therefore be independent standard normal variables. In practice parameters are estimated, so (A-09) is a diagnostic reference rather than an exact finite-sample distribution. A normal quantile plot can reveal strong skewness, heavy tails or outliers, but it cannot by itself prove model adequacy.

Projected geometry policy

For point `sf` input, SPmodels extracts the two geometry coordinates only after checking that:

1. every geometry is a non-empty two-dimensional point;
2. a coordinate reference system is known;
3. the CRS is projected rather than longitude/latitude;
4. prediction data use the same CRS as the fitting data.

The package does not guess a projection or silently transform prediction data. For ordinary data frames, the user supplies two coordinate columns and states their shared unit. This policy makes every distance in (A-04)–(A-07) explicit.

Algorithms and code map

Operation	Algorithm	Implementation
Response simulation	A-01 using the stored Cholesky factor	`simulate.spmodels_fit()`
Pair semivariance	A-04 for the lower triangle	`sp_variogram()`
Bin estimate	A-05 with equal-width bins and counts	`sp_variogram()`
Fitted variogram	A-07 at bin midpoints	`sp_variogram()`
Decorrelated residual	A-08 by triangular solve	`sp_diagnostics()`
Geometry checks	Explicit projected-point policy	`.sp_coordinates()`

Independent validation

The analytic tests reconstruct simulations from the identical seeded normal draws and stored Cholesky factor, verify that every unordered pair is counted by the variogram, and reconstruct (A-08) directly. Geometry tests fit and predict with projected point data and reject longitude/latitude data, duplicate coordinate specifications and mismatched prediction CRSs.

These tests validate the stated algorithms. Simulation assessment across many replicates and broader reference comparisons remain future work. Formal permutation and fitted-model variogram envelopes are derived separately in derivation 11.

Matérn scale derivatives and non-Gaussian latent modes

Why these operations are needed

Binomial and Poisson geostatistical regression introduce a latent Gaussian spatial vector inside a nonlinear response model. Later parameter estimation needs covariance derivatives, a stable conditional latent mode and the curvature at that mode. This chapter derives those deterministic building blocks. It does not claim that non-Gaussian model fitting is complete.

Matérn notation

Write the SPmodels Matérn correlation as

$$R(d;\rho,\nu) = C_\nu x^\nu K_\nu(x), \qquad C_\nu = \frac{2^{1-\nu}}{\Gamma(\nu)}, \qquad x = \frac{d}{\rho}. \tag{N-01}$$

Distance d and smoothness ν are held fixed while differentiating with respect to scale ρ. Since

$$\frac{dx}{d\rho} = -\frac{x}{\rho}, \tag{N-02}$$

the scale derivative follows from the Bessel identity

$$\frac{d}{dx}\{x^\nu K_\nu(x)\} = -x^\nu K_{\nu-1}(x). \tag{N-03}$$

Combining (N-01)–(N-03),

$$\boxed{\frac{\partial R}{\partial \rho} = \frac{C_\nu}{\rho} x^{\nu+1} K_{\nu-1}(x) = \frac{R}{\rho} x \frac{K_{\nu-1}(x)}{K_\nu(x)}.} \tag{N-04}$$

At $d = 0$, correlation is exactly one for every scale, so both scale derivatives are zero.

Second scale derivative

Differentiate (N-04), using

$$\frac{dK_a(x)}{dx} = -\frac{1}{2}\{K_{a-1}(x) + K_{a+1}(x)\} \tag{N-05}$$

and the Bessel recurrence relation. After collecting terms,

$$\boxed{\frac{\partial^2 R}{\partial \rho^2} = \frac{R}{\rho^2}\left[x^2 - (2\nu+1)x\frac{K_{\nu-1}(x)}{K_\nu(x)}\right].} \tag{N-06}$$

For $\nu = 1/2$, $R = \exp(-d/\rho)$, so the independently derived closed forms are

$$\frac{\partial R}{\partial \rho} = R\frac{d}{\rho^2}, \qquad \frac{\partial^2 R}{\partial \rho^2} = R\left(\frac{d^2}{\rho^4} - \frac{2d}{\rho^3}\right). \tag{N-07}$$

These provide sign-sensitive analytic tests. General smoothness values are also compared with centred finite differences using predeclared tolerances. The implementation evaluates ratios of exponentially scaled Bessel functions, so their common exponential factor cancels.

Non-Gaussian latent model

Let s be a latent Gaussian vector with

$$s \sim N(0, \Sigma), \qquad \eta = a + s, \tag{N-08}$$

where $a = X\beta + o$ is supplied to the mode calculation. Conditional on s, observations are independent.

For binomial-logit data,

$$Y_i \mid s_i \sim \text{Binomial}(m_i, p_i), \qquad p_i = \frac{1}{1+\exp(-\eta_i)}. \tag{N-09}$$

For Poisson-log data,

$$Y_i \mid s_i \sim \text{Poisson}(\mu_i), \qquad \mu_i = \exp(\eta_i). \tag{N-10}$$

Ignoring constants that do not depend on s, the latent log posterior is

$$h(s) = \sum_i \ell_i(\eta_i) - \frac{1}{2}s^T\Sigma^{-1}s. \tag{N-11}$$

Score and curvature

For both supported families, define the conditional mean vector μ. The likelihood score with respect to η is $y - \mu$. Therefore

$$\boxed{g(s) = \frac{\partial h}{\partial s} = y - \mu - \Sigma^{-1} s.} \tag{N-12}$$

Let W be diagonal. Its entries are

$$W_{ii} = m_i p_i (1 - p_i) \quad \text{for binomial data,} \tag{N-13}$$

and

$$W_{ii} = \mu_i \quad \text{for Poisson data.} \tag{N-14}$$

The negative Hessian is

$$\boxed{H(s) = -\frac{\partial^2 h}{\partial s \partial s^T} = W + \Sigma^{-1}.} \tag{N-15}$$

Because Σ^{-1} is positive definite and W is non-negative, $H(s)$ is positive definite. The log posterior is consequently strictly concave and has one mode.

Damped Newton algorithm

Starting from zero or a supplied vector, solve

$$H(s^{(k)}) \Delta^{(k)} = g(s^{(k)}) \tag{N-16}$$

by Cholesky factorisation, then propose

$$s^{(k+1)} = s^{(k)} + \alpha \Delta^{(k)}, \tag{N-17}$$

where α begins at one. If (N-17) decreases (N-11), halve α until the objective does not decrease. This damping protects against an over-large Newton step when the starting point is far from the mode.

Convergence is declared when

$$\max_i |\alpha \Delta_i| \le \epsilon \{1 + \max_i |s_i|\}. \tag{N-18}$$

At the final point, the returned local Gaussian covariance is

$$\boxed{C_{\text{mode}} = H(\hat{s})^{-1}.} \tag{N-19}$$

This covariance describes local curvature. Returning it does not turn the calculation into Bayesian parameter estimation and does not supply MCML by itself.

Metropolis-adjusted Langevin sampling

Let $C = H(\hat{s})^{-1}$ be the fixed mode-based preconditioner from (N-19), let $g(s)$ be (N-12), and choose step size $\delta > 0$. From current state s, MALA proposes

$$\boxed{s' = s + \frac{\delta^2}{2} C g(s) + \delta C^{1/2} z, \qquad z \sim N(0, I).} \tag{N-20}$$

Thus the forward proposal density is

$$q(s' \mid s) = N\left(s';\ s + \frac{\delta^2}{2} C g(s),\ \delta^2 C\right). \tag{N-21}$$

Because the drift depends on the current state, this proposal is asymmetric. The Metropolis correction is essential. Accept s' with probability

$$\boxed{\alpha(s, s') = \min\left\{1, \frac{\exp\{h(s')\}\, q(s \mid s')}{\exp\{h(s)\}\, q(s' \mid s)}\right\}.} \tag{N-22}$$

SPmodels evaluates (N-22) on the log scale. The common proposal determinant cancels because C and δ are fixed. For proposal mean $m(s) = s + \delta^2 C g(s)/2$, the remaining forward quadratic is

$$Q(s', s) = \frac{1}{\delta^2}\{s' - m(s)\}^T C^{-1}\{s' - m(s)\}. \tag{N-23}$$

The log acceptance ratio is consequently

$$h(s') - h(s) - \frac{1}{2} Q(s, s') + \frac{1}{2} Q(s', s). \tag{N-24}$$

The sampler starts at the latent mode, discards the declared burn-in and keeps every declared thinning interval. A supplied random seed is local to the call. The result reports the realised acceptance rate; it does not automatically retune a poorly chosen step size or certify convergence.

Autocorrelation and effective sample size

Retained MCMC draws are dependent. For one latent component, let ρ_k denote the sample autocorrelation at lag k. If N draws are retained, the ideal infinite-series definition of integrated autocorrelation time is

$$\tau_{\text{int}} = 1 + 2\sum_{k=1}^{\infty} \rho_k. \tag{N-25}$$

The corresponding effective sample size is

$$\boxed{N_{\text{eff}} = \frac{N}{\tau_{\text{int}}}.} \tag{N-26}$$

An infinite noisy sum is not usable directly. SPmodels groups the estimated autocorrelations into adjacent pairs,

$$P_j = \hat{\rho}_{2j-1} + \hat{\rho}_{2j}, \tag{N-27}$$

and stops immediately before the first non-positive pair. If J positive pairs remain, it uses

$$\hat{\tau}_{\text{int}} = \max\left\{1, 1 + 2\sum_{k=1}^{2J}\hat{\rho}_k\right\}, \qquad \widehat{N}_{\text{eff}} = \min\left\{N, \frac{N}{\hat{\tau}_{\text{int}}}\right\}. \tag{N-28}$$

For one chain, `sp_sampler_diagnostics()` reports this estimate for every latent component and trace and autocorrelation values for one selected component. A satisfactory trace, acceptance rate or effective sample size is not proof of convergence, which motivates the multiple-chain extension below.

Multiple chains and dispersed initial states

Let M be the requested number of chains. Each chain uses the same posterior and MALA transition but evolves with its own sequence of random proposals and acceptance uniforms. When the caller does not supply starting values, chain one starts at the latent mode $\hat{s}$. For chains $m = 2, \dots, M$, SPmodels draws

$$s_0^{(m)} = \hat{s} + c_0 L z_m, \qquad z_m \sim N(0, I), \tag{N-29}$$

where $LL^{\mathsf{T}} = P$ and c_0 is `initial_scale`. Thus $c_0 = 0$ starts every chain at the mode, whereas a positive value deliberately tests whether chains started in different posterior regions mix towards the same distribution. A supplied matrix instead defines every initial state explicitly.

The returned three-dimensional array keeps retained draw, latent component and chain as separate dimensions. A pooled matrix is also returned for backwards compatibility, but between-chain diagnostics always use the chain array.

Rank-normalized split R-hat

For one scalar component, split each of the M chains into its first and last halves, discarding the middle draw when the retained length is odd. This gives $2M$ sequences of length n. Rank all $2Mn$ values jointly. For pooled rank r, use the normal score

$$z = \Phi^{-1}\left(\frac{r - 3/8}{2Mn + 1/4}\right). \tag{N-30}$$

For the normalized split sequences, let $\bar{z}_m$ and s_m^2 be the chain mean and sample variance. The average within-chain variance and scaled between-chain variance are

$$W = \frac{1}{2M}\sum_{m=1}^{2M} s_m^2, \qquad B = n\,\mathrm{Var}(\bar{z}_1, \dots, \bar{z}_{2M}). \tag{N-31}$$

Estimate the marginal variance by

$$\widehat{V}^{+} = \frac{n-1}{n}W + \frac{B}{n}, \qquad \widehat{R} = \sqrt{\frac{\widehat{V}^{+}}{W}}. \tag{N-32}$$

Rank normalization makes this comparison less sensitive to a skewed scale. Location agreement alone can miss chains with different spreads, so SPmodels also folds the original draws around their pooled median, rank-normalizes the absolute deviations, and repeats equations N-31–N-32. The reported value is

$$\widehat{R}_{\text{reported}} = \max(\widehat{R}_{\text{rank}}, \widehat{R}_{\text{folded}}). \tag{N-33}$$

A value near one is necessary but not sufficient for convergence. SPmodels records whether every component is at most 1.01 but does not discard or alter draws to force that result. R-hat is unavailable for a single selected chain.

Multi-chain ESS and Monte Carlo standard error

For retained chain m, centre draws at their chain mean and estimate the lag t autocovariance using divisor n. Average it over chains to obtain $\hat{\gamma}_t$. Using W and $\widehat{V}^+$ on the original draw scale, estimate pooled autocorrelation by

$$\hat{\rho}_t = 1 - \frac{W - \hat{\gamma}_t}{\widehat{V}^+}. \tag{N-34}$$

SPmodels groups adjacent autocorrelations as in N-27, truncates before the first non-positive pair, and makes the retained pair sums non-increasing using cumulative minima. With integrated autocorrelation estimate $\hat{\tau}_{\text{int}}$, the multi-chain effective sample size is

$$\widehat{N}_{\text{eff}} = \min\left\{Mn, \frac{Mn}{\hat{\tau}_{\text{int}}}\right\}. \tag{N-35}$$

Finally, if s is the standard deviation of all pooled retained draws, the Monte Carlo standard error of their mean is

$$\boxed{\text{MCSE}(\bar{s}) = \frac{s}{\sqrt{\widehat{N}_{\text{eff}}}}.} \tag{N-36}$$

MCSE describes simulation error in a posterior mean conditional on the supplied model parameters. It is not the posterior standard deviation and not a standard error for estimated regression or covariance parameters.

This is conditional latent sampling for supplied parameters. Sampling latent effects does not make regression or covariance parameter estimation Bayesian, and it is not yet the Monte Carlo maximum-likelihood estimator required by the full non-Gaussian model.

Input and failure conditions

The covariance must be finite, symmetric and positive definite. Binomial successes and trials must be non-negative integers with $y_i \leq m_i$. Poisson outcomes must be non-negative integers. Only logit and log links are accepted. Overflowing proposals are rejected during step halving; unusable curvature raises an error rather than being silently modified.

Code and validation map

Equations	Implementation	Independent checks
N-04, N-06	`matern_scale_derivative()`	Exponential identities and centred finite differences
N-09–N-15	`.sp_latent_components()`	Direct scores and curvature matrices
N-16–N-19	`sp_latent_mode()`	Near-zero final score, inverse curvature and row permutation
N-20–N-24	`sp_latent_sample()`	Seed reproducibility and moments against numerical integration
N-25–N-28	`sp_sampler_diagnostics()`	Single-chain ESS bounds and requested ACF length
N-29	`sp_latent_sample()`	Local seeded reproducibility, dispersed starts and retained chain identity
N-30–N-36	`sp_sampler_diagnostics()`	Direct MCSE identity, multi-chain ESS bounds, chain selection and rank-normalized split R-hat

The relevant tests are in `tests/test-nongaussian-foundation.R`. The one-dimensional binomial posterior mean and standard deviation are checked against numerical integration using predeclared absolute margins. Four seeded chains test reproducibility, chain identities, dispersed initialization, multi-chain ESS/MCSE, chain selection and a converged rank-normalized split R-hat. External reference comparisons remain pending.

Binomial and Poisson geostatistical fitting by Laplace approximation

Scope and assumptions

This chapter derives the first fitted non-Gaussian models in SPmodels. It is important to name the approximation: this implementation maximises a **Laplace approximation** to the marginal likelihood. It is not Monte Carlo maximum likelihood.

The observations are point supported. Conditional on a latent Gaussian vector, they are independent. Coordinates lie in a two-dimensional Euclidean system with declared units. The mean-model design matrix has full column rank. Matérn smoothness is fixed.

Linear predictor and covariance

For observation i,

$$\eta_i = x_i^\mathsf{T}\beta + o_i + s_i, \tag{L-01}$$

where o_i is a known offset. The latent vector has distribution

$$s \sim N(0, \Sigma), \tag{L-02}$$

with entries

$$\Sigma_{ij} = \sigma_s^2 R(d_{ij}; \rho, \nu) + \sigma_e^2 \mathbb{1}(i = j). \tag{L-03}$$

Here $\sigma_s^2 > 0$ is spatial variance, $\rho > 0$ is Matérn scale, fixed $\nu > 0$ is smoothness, and $\sigma_e^2 > 0$ is an independent latent-effect variance. Unlike Gaussian measurement noise, this independent effect enters before the inverse link.

Response likelihoods

For binomial-logit data,

$$Y_i \mid s_i \sim \text{Binomial}(m_i, p_i), \qquad p_i = \{1 + \exp(-\eta_i)\}^{-1}. \tag{L-04}$$

SPmodels accepts successes and failures as a two-column response, a proportion with integer trial weights, or a zero-one Bernoulli response. Each encoding is converted to successes y_i and trials m_i, so it defines the same log likelihood

$$\ell_B(s) = \sum_i \left[\log \binom{m_i}{y_i} + y_i \log p_i + (m_i - y_i) \log(1 - p_i) \right]. \tag{L-05}$$

For Poisson-log data,

$$Y_i \mid s_i \sim \text{Poisson}(\mu_i), \qquad \mu_i = \exp(\eta_i), \tag{L-06}$$

and

$$\ell_P(s) = \sum_i \{y_i \log \mu_i - \mu_i - \log(y_i!)\}. \tag{L-07}$$

Known exposure belongs in $o_i = \log(e_i)$. Generic Poisson weights are not silently reinterpreted as exposure.

The marginal likelihood problem

Let $\theta = (\beta, \sigma_s^2, \rho, \sigma_e^2)$, and write ℓ_y for (L-05) or (L-07). The exact marginal likelihood integrates out the latent vector:

$$L(\theta) = \int \exp\{\ell_y(s)\} \phi_n(s; 0, \Sigma)\, ds. \tag{L-08}$$

For nonlinear links this integral generally has no closed form. Define

$$h(s) = \ell_y(s) - \frac{1}{2} s^\top \Sigma^{-1} s. \tag{L-09}$$

Chapter 4 derives the unique mode $\hat{s}$ and negative Hessian

$$H = \Sigma^{-1} + W \tag{L-10}$$

at that mode.

Laplace approximation

A second-order Taylor expansion around the mode is

$$h(s) \approx h(\hat{s}) - \frac{1}{2}(s-\hat{s})^{\mathsf{T}} H (s-\hat{s}). \tag{L-11}$$

Integrating the Gaussian quadratic in (L-11) and retaining every normalising term that depends on the parameters gives

$$\boxed{\tilde{\ell}(\theta) = \ell_y(\hat{s}) - \frac{1}{2}\hat{s}^{\mathsf{T}}\Sigma^{-1}\hat{s} - \frac{1}{2}\log|\Sigma| - \frac{1}{2}\log|H|.} \tag{L-12}$$

The $n\log(2\pi)/2$ term from the Gaussian prior cancels the corresponding Laplace integration term. Since the latent-mode routine returns $C = H^{-1}$, the implementation evaluates

$$-\frac{1}{2}\log|H| = \frac{1}{2}\log|C|. \tag{L-13}$$

All determinants and linear solves use Cholesky factors; the implementation does not form a determinant by multiplying eigenvalues.

Parameter optimisation

The regression coefficients are initialised by an independent generalized linear-model calculation. Positive covariance parameters are represented on the log scale:

$$\gamma = (\log\sigma_s^2, \log\rho, \log\sigma_e^2). \tag{L-14}$$

At every trial value (β, γ), SPmodels:

1. constructs Σ;
2. finds $\hat{s}$ by damped Newton iteration;
3. evaluates (L-12); and
4. returns its negative to a bounded optimiser.

If covariance parameters are fixed, only β is optimised. Otherwise all components are optimised jointly with fixed smoothness. If a trial point cannot produce a positive-definite covariance, converged latent mode or finite likelihood, the objective returns the finite sentinel 10^{20}. A finite sentinel keeps bounded numerical differentiation from overflowing when one side of a trial step is invalid; it is far above any accepted objective in the supported data sizes. The final parameters must still produce a usable state, and non-zero optimizer convergence is reported rather than hidden.

Reported uncertainty and fitted values

When the numerical Hessian of the approximated negative log likelihood is positive definite, its inverse supplies a local parameter covariance matrix:

$$\widehat{\mathrm{Var}}\begin{pmatrix}\hat{\beta}\\ \hat{\gamma}\end{pmatrix} = \left\{\nabla^2[-\tilde{\ell}]\right\}^{-1}. \tag{L-15}$$

The delta method transforms the covariance-parameter block back from log scale, as in the Gaussian derivation. If curvature is not usable, standard errors are reported as missing rather than repaired silently.

Fitted values are conditional response means at the latent mode:

$$\hat{y}_i = \begin{cases} m_i \operatorname{logit}^{-1}(x_i^\mathsf{T}\hat{\beta} + o_i + \hat{s}_i), & \text{binomial,} \\ \exp(x_i^\mathsf{T}\hat{\beta} + o_i + \hat{s}_i), & \text{Poisson.} \end{cases} \tag{L-16}$$

They are not marginal means integrated over latent uncertainty.

Point prediction under the Laplace approximation

Let u_* be a latent effect at a new point and let c_* contain its spatial covariances with the fitted sites. The independent latent component contributes no cross-covariance at a new site. Conditional Gaussian identities give

$$u_* \mid s \sim N\left(c_*^\mathsf{T}\Sigma^{-1}s, v_0 - c_*^\mathsf{T}\Sigma^{-1}c_*\right), \tag{L-17}$$

where $v_0 = \sigma_s^2$ for the smooth latent target and $v_0 = \sigma_s^2 + \sigma_e^2$ for a new observation-site target. Combining (L-17) with the Laplace posterior $s \mid y \approx N(\hat{s}, C)$ yields an approximate normal predictive linear predictor with mean

$$m_* = x_*^\mathsf{T}\hat{\beta} + o_* + c_*^\mathsf{T}\Sigma^{-1}\hat{s} \tag{L-18}$$

and variance

$$v_* = v_0 - c_*^\mathsf{T}\Sigma^{-1}c_* + c_*^\mathsf{T}\Sigma^{-1}C\Sigma^{-1}c_*. \tag{L-19}$$

The last term propagates local latent-mode uncertainty. Covariance-parameter and regression-parameter uncertainty are not included.

For a Poisson model, lognormal moments give the marginal intensity

$$E(\mu_* \mid y) = \exp(m_* + v_*/2), \tag{L-20}$$

and its variance

$$\operatorname{Var}(\mu_* \mid y) = E(\mu_* \mid y)^2\{\exp(v_*) - 1\}. \tag{L-21}$$

For a future Poisson count, conditional sampling variation adds $E(\mu_* \mid y)$ to (L-21).

For a binomial model, there is no corresponding elementary logistic-normal mean. SPmodels evaluates

$$\overline{p}_* = \int_{-\infty}^{\infty} \operatorname{logit}^{-1}(m_* + \sqrt{v_*}z)\phi(z)\,dz \tag{L-22}$$

and

$$E(p_*^2 \mid y) = \int_{-\infty}^{\infty} \{\operatorname{logit}^{-1}(m_* + \sqrt{v_*}z)\}^2\phi(z)\,dz \tag{L-23}$$

by numerical quadrature. Their difference gives latent probability variance. For a future Bernoulli outcome the predictive variance simplifies to

$$\mathrm{Var}(Y_* \mid y) = \overline{p}_*(1 - \overline{p}_*). \tag{L-24}$$

Thus response prediction integrates through the nonlinear link; it does not merely apply the inverse link to (L-18).

Limitations and validation map

Laplace accuracy can be poor for sparse binary data, small cluster sizes, large latent variances or weakly identified covariance parameters. Optimiser convergence does not establish approximation accuracy. Later chapters derive fitted-model simulation, conditional residual diagnostics, MCML correction and optional Hessian-based parameter uncertainty in joint prediction. Those extensions do not alter the preliminary Laplace equations in this chapter.

Equations	Implementation	Independent checks
L-01–L-07	`.sp_response_data()` and model-frame construction	Equivalent binomial encodings, Poisson offset and invalid-input tests
L-08–L-13	`.sp_laplace_state()`	Direct reconstruction of all four terms in (L-12)
L-14–L-15	`.sp_fit_nongaussian()`	Positivity, convergence reporting and explicit missing-curvature policy
L-16	`fitted.spmodels_fit()`	Family-specific range checks
L-17–L-24	`.sp_predict_nongaussian()`	Direct Poisson lognormal moments and binomial numerical integration

Row-permutation invariance is checked with a simultaneous permutation of responses, predictors, and coordinates. Reference comparison and stochastic full reference validation remains separate future work.

Joint prediction, grids and draw-based targets

Why marginal standard errors are insufficient

A set of pointwise prediction means and standard errors does not describe how prediction errors move together. Regional averages, maxima and exceedance summaries depend on the joint distribution. SPmodels therefore constructs one predictive mean vector and covariance matrix, draws all target locations together, and applies transformations or aggregation separately to each joint draw.

By default, results in this chapter are conditional on fitted covariance parameters. Section J-15–J-18 adds the optional local Hessian approximation used when `parameter_uncertainty = TRUE`. The non-Gaussian results additionally inherit the Laplace approximation.

Gaussian joint prediction

Let y be observed Gaussian responses with covariance V. At m new locations, let K_{**} be the $m \times m$ spatial covariance matrix and C the $n \times m$ covariance between observed and new locations. Define

$$\hat{\beta} = (X^{\mathsf{T}} V^{-1} X)^{-1} X^{\mathsf{T}} V^{-1} (y - o), \tag{J-01}$$

and

$$Q = X_* - C^{\mathsf{T}} V^{-1} X. \tag{J-02}$$

The joint universal-kriging mean is

$$\boxed{m_* = X_* \hat{\beta} + o_* + C^{\mathsf{T}} V^{-1} (y - o - X\hat{\beta}).} \tag{J-03}$$

The covariance for the smooth latent target is

$$\boxed{S_* = K_{**} - C^{\mathsf{T}} V^{-1} C + Q (X^{\mathsf{T}} V^{-1} X)^{-1} Q^{\mathsf{T}}.} \tag{J-04}$$

The first subtraction is uncertainty reduction from the observations. The last term accounts for estimating the regression mean while treating covariance parameters as fixed. For a future noisy observation, add nugget variance to every diagonal entry:

$$S_{*,\text{obs}} = S_* + \tau^2 I_m. \tag{J-05}$$

For a non-spatial Gaussian fit, (J-03) reduces to the regression mean and (J-04) reduces to $X_* \widehat{\text{Var}}(\hat{\beta}) X_*^{\mathsf{T}}$.

Non-Gaussian joint prediction

For binomial and Poisson fits, Chapter 5 gives the Laplace posterior

$$s \mid y \mathrel{\dot{\sim}} N(\hat{s}, C_s). \tag{J-06}$$

Let Σ be the fitted latent covariance at observed sites, C the spatial cross-covariance, and K_{**} the new-site spatial covariance. Applying total expectation and total covariance gives predictive link mean

$$\boxed{m_* = X_* \hat{\beta} + o_* + C^{\mathsf{T}} \Sigma^{-1} \hat{s}} \tag{J-07}$$

and covariance

$$\boxed{S_* = K_{**} - C^{\mathsf{T}} \Sigma^{-1} C + C^{\mathsf{T}} \Sigma^{-1} C_s \Sigma^{-1} C.} \tag{J-08}$$

The last term propagates local latent posterior uncertainty. Adding $\sigma_e^2 I_m$ produces the new observation-site link covariance. Regression- and covariance-parameter uncertainty are not included in the default conditional calculation; the optional approximation is derived below.

Generating joint draws

For either model class, let S_* denote the selected predictive covariance and choose a matrix A satisfying $AA^{\mathsf{T}} = S_*$. For draw b,

$$z^{(b)} \sim N(0, I_m), \qquad \eta_*^{(b)} = m_* + Az^{(b)}. \tag{J-09}$$

SPmodels uses a Cholesky factor when the covariance is positive definite. If rounding leaves a positive-semidefinite matrix with tiny zero or negative eigenvalues, it checks those values against a scale-dependent tolerance and sets only tolerable negative values to zero. A materially negative eigenvalue raises an error.

Gaussian response and link draws coincide. For non-Gaussian latent response targets,

$$p_*^{(b)} = \text{logit}^{-1}(\eta_*^{(b)}) \tag{J-10}$$

or

$$\mu_*^{(b)} = \exp(\eta_*^{(b)}). \tag{J-11}$$

For future observations, SPmodels adds response sampling:

$$Y_{*i}^{(b)} \sim \text{Bernoulli}(p_{*i}^{(b)}) \tag{J-12}$$

or

$$Y_{*i}^{(b)} \sim \text{Poisson}(\mu_{*i}^{(b)}). \tag{J-13}$$

A supplied seed is local to the function and does not alter the caller's random-number stream.

Optional fitted-parameter uncertainty

The default draws above condition on fitted regression and covariance parameters. When a usable observed-Hessian covariance is available, SPmodels can add a local asymptotic parameter layer. Let

$$\psi = (\beta^{\mathsf{T}}, \log \sigma_s^2, \log \rho, \log \sigma_e^2)^{\mathsf{T}}. \tag{J-14}$$

For a non-Gaussian Laplace fit, the joint numerical Hessian supplies $\widehat{V}_\psi$. The approximation is

$$\psi^{(b)} \sim N(\widehat{\psi}, \widehat{V}_\psi). \tag{J-15}$$

For a Gaussian fit, the available profile covariance and conditional regression covariance do not contain their cross-covariance. SPmodels makes the explicit approximation

$$\widehat{V}_\psi = \begin{pmatrix} \widehat{\text{Var}}(\hat{\beta}) & 0 \\ 0 & \widehat{\text{Var}}(\hat{\gamma}) \end{pmatrix}, \tag{J-16}$$

where γ contains log covariance parameters and its block is obtained by the inverse delta method. This block-independence assumption is reported rather than hidden.

For every parameter draw, covariance parameters are transformed back to the positive scale. SPmodels reconstructs the Gaussian conditioning system or the non-Gaussian latent mode, then generates one draw using (J-09). Consequently, parameter and conditional predictive uncertainty are combined in the final draw collection. An unusable Hessian covariance or a parameter draw that cannot produce a valid conditioning system raises an error. Draws are not silently truncated or repaired.

Weighted and grouped targets

Suppose locations assigned to region g have non-negative raw weights a_{ig}. SPmodels requires positive total weight in every region and uses

$$w_{ig} = \frac{a_{ig}}{\sum_{j \in g} a_{jg}}. \tag{J-17}$$

The regional target for draw b is calculated before summarisation:

$$T_g^{(b)} = \sum_{i \in g} w_{ig} D_i^{(b)}, \tag{J-18}$$

where $D_i^{(b)}$ is the requested point-level draw. This order retains spatial covariance. Combining marginal variances after the fact would not.

The reported Monte Carlo mean and standard deviation are

$$\overline{T}_g = \frac{1}{B} \sum_{b=1}^{B} T_g^{(b)}, \qquad s_g = \left\{ \frac{1}{B-1} \sum_{b=1}^{B} (T_g^{(b)} - \overline{T}_g)^2 \right\}^{1/2}. \tag{J-19}$$

Quantiles are calculated directly from $\{T_g^{(b)}\}$. For threshold c, the exceedance estimate is

$$\widehat{P}(T_g > c) = \frac{1}{B} \sum_{b=1}^{B} \mathbb{1}\{T_g^{(b)} > c\}. \tag{J-20}$$

This is Monte Carlo uncertainty conditional on the fitted model; it is not a frequentist confidence interval for the fitted parameters.

Projected grid construction

`sp_grid()` operates only in a known projected CRS so that cell sizes have the declared coordinate units. Its algorithm is:

1. validate a non-empty projected `sf` object and positive x/y spacing;
2. use the union of polygon inputs, an explicit polygon boundary, or the convex hull of point inputs;
3. reject non-polygonal or zero-area boundaries;
4. construct regular cell centres over the boundary bounding box;
5. retain centres intersecting the boundary; and
6. order retained points from top to bottom and then left to right, assigning stable `grid_id` values.

The function does not guess a projection, transform longitude/latitude, or claim that a centre-point grid performs exact polygon integration.

Point-to-polygon allocation

`sp_polygon_weights()` assigns each projected prediction point to a projected polygon region using strict point-within-polygon membership. It requires the same CRS, non-missing region identifiers and finite non-negative point weights. Points assigned to multiple distinct region identifiers raise an error. Points outside every polygon either raise an error or are explicitly dropped, as selected by the caller.

The resulting indices, region identifiers and raw weights feed directly into (J-17)–(J-20). This is centre-point allocation. It is appropriate for point support or a grid-centre approximation. Exact fractional cell-area overlay for coarse cells that cross regional boundaries is derived in derivation 11.

Code and validation map

Equations or algorithm	Implementation	Independent checks
J-01–J-05	`.sp_joint_gaussian()`	Direct universal-kriging mean, complete covariance and seeded matrix transform
J-06–J-08	`.sp_joint_nongaussian()`	Marginal variances equal pointwise prediction variances
J-09–J-13	`sp_predict_draws()`	Local seed, support checks and exact seeded Gaussian transform
J-14–J-16	parameter layer in `sp_predict_draws()`	Local reproducibility, recorded draws and family-specific fitted-state reconstruction
J-17–J-20	`sp_predict_target()`	Direct weighted draws, normalisation, quantiles and exceedance frequencies
Grid algorithm	`sp_grid()`	CRS rejection, convex hull, stable ordering and degenerate-boundary failure
Polygon allocation	`sp_polygon_weights()`	Direct regional weighted reconstruction and CRS mismatch failure

Exact bounded-memory pointwise grid prediction is derived in Chapter 18. Chapter 19 derives the optional bounded-memory nearest-predecessor approximation for joint draws; ordinary `sp_predict_draws()` remains the exact route for the declared fitted predictive distribution. Exact fractional cell-area weights are implemented and derived in derivation 11.

Cross-validation and predictive-draw assessment

Purpose and separation of tasks

Cross-validation measures prediction after withholding observations and refitting. Predictive-draw assessment compares an already generated ensemble with known simulated truth or held-out values. These are different operations: the first guards against training-data leakage, while the second evaluates calibration and accuracy of a predictive distribution.

Fixed-fold refitting

Let $f_i \in \{1, \dots, K\}$ identify the fold for observation i. For fold k, define training and test index sets

$$\mathcal{T}_k = \{i : f_i \neq k\}, \qquad \mathcal{H}_k = \{i : f_i = k\}. \tag{V-01}$$

SPmodels refits the complete declared model using only $\mathcal{T}_k$, then predicts observations in $\mathcal{H}_k$. Covariance parameters are therefore re-estimated when the original control requested estimation; they are not carried over from the full-data fit. Fixed covariance controls remain fixed.

User-supplied fold identifiers are preserved. If only a fold count is given, SPmodels randomly permutes a balanced repetition of $1, \dots, K$. A supplied seed is local to fold construction. Random folds are a reproducibility convenience, not a substitute for scientifically justified spatial blocking.

Held-out response scale

For Gaussian and Poisson models, the held-out value is y_i. For a binomial response with y_i successes from m_i trials, response-scale error uses

$$r_i = \frac{y_i}{m_i} - \hat{p}_i. \tag{V-02}$$

For other supported families,

$$r_i = y_i - \hat{\mu}_i. \tag{V-03}$$

Across N held-out predictions, root mean squared error and mean absolute error are

$$\mathrm{RMSE} = \left(\frac{1}{N} \sum_{i=1}^{N} r_i^2 \right)^{1/2}, \qquad \mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} |r_i|. \tag{V-04}$$

These are reported overall and separately by fold.

Negative log score

The negative log score penalises low probability assigned to what occurred. For a Gaussian held-out predictive mean $\hat{\mu}_i$ and standard deviation $\hat{\sigma}_i$,

$$S_i = -\log \phi(y_i; \hat{\mu}_i, \hat{\sigma}_i^2). \tag{V-05}$$

For a non-Gaussian held-out row, the local predictive approximation on the link scale is

$$\eta_i \mid Y_{\mathcal{T}_k} \mathrel{\dot{\sim}} N(m_i^*, v_i^*). \tag{V-06}$$

For a binomial observation with n_i trials, let a_i be any fitted intervention multiplier (one when absent). The predictive mass is

$$q_i(y_i) = \int \Pr\{Y_i = y_i \mid n_i, a_i \, \mathrm{logit}^{-1}(\eta)\} \phi(\eta; m_i^*, v_i^*) \, d\eta. \tag{V-07}$$

For Poisson data it is

$$q_i(y_i) = \int \Pr\{Y_i = y_i \mid \exp(\eta)\}\phi(\eta; m_i^*, v_i^*)\, d\eta. \tag{V-08}$$

SPmodels evaluates each one-dimensional integral adaptively after writing $\eta = m_i^* + \sqrt{v_i^*}z$, $z \sim N(0,1)$, and reports $S_i = -\log q_i(y_i)$. A failed or underflowed integral is an explicit error, not a clipped finite score. The reported mean is $N^{-1}\sum_i S_i$; smaller is better. Integration is conditional on fitted parameters and the local Gaussian predictive approximation.

Assessment from predictive draws

Suppose $D_i^{(1)}, \ldots, D_i^{(B)}$ are predictive draws for location i and t_i is known truth. The draw mean and error are

$$\overline{D}_i = \frac{1}{B}\sum_{b=1}^{B} D_i^{(b)}, \qquad e_i = \overline{D}_i - t_i. \tag{V-09}$$

Bias, RMSE and MAE aggregate these errors across locations:

$$\text{Bias} = \frac{1}{M}\sum_i e_i, \qquad \text{RMSE} = \left(\frac{1}{M}\sum_i e_i^2\right)^{1/2}, \qquad \text{MAE} = \frac{1}{M}\sum_i |e_i|. \tag{V-10}$$

For central probability $1-\alpha$, empirical draw quantiles define $[L_i, U_i]$. Coverage and mean width are

$$\text{Coverage} = \frac{1}{M}\sum_i \mathbb{1}(L_i \le t_i \le U_i), \qquad \text{Width} = \frac{1}{M}\sum_i (U_i - L_i). \tag{V-11}$$

Empirical continuous ranked probability score

For one location, the empirical CRPS is

$$\boxed{\text{CRPS}_i = \frac{1}{B}\sum_{b=1}^{B} |D_i^{(b)} - t_i| - \frac{1}{2B^2}\sum_{b=1}^{B}\sum_{c=1}^{B} |D_i^{(b)} - D_i^{(c)}|.} \tag{V-12}$$

It rewards both calibration and concentration; smaller is better. Directly forming all B^2 differences is unnecessary. If $D_{i(1)} \le \cdots \le D_{i(B)}$ are sorted draws, then

$$\frac{1}{2B^2}\sum_{b,c} |D_i^{(b)} - D_i^{(c)}| = \frac{1}{B^2}\sum_{b=1}^{B} (2b - B - 1) D_{i(b)}. \tag{V-13}$$

SPmodels uses (V-12) and reports the mean CRPS across locations.

For optional threshold q, it also reports

$$\widehat{P}(D_i > q) = \frac{1}{B}\sum_b \mathbb{1}(D_i^{(b)} > q) \tag{V-14}$$

alongside $\mathbb{1}(t_i > q)$. No arbitrary classification cutoff is imposed.

Calibration curve and finite-location bounds

One coverage value at one chosen interval level cannot show whether predictive intervals are systematically too narrow or too wide. For each requested nominal central coverage p, SPmodels calculates the type-8 draw quantiles at $(1-p)/2$ and $(1+p)/2$. If these are $L_i(p)$ and $U_i(p)$, the empirical calibration curve is

$$\hat{c}(p) = \frac{1}{M}\sum_{i=1}^{M} \mathbb{1}\{L_i(p) \le t_i \le U_i(p)\}. \tag{V-15}$$

A calibrated ensemble should lie near the identity line $\hat{c}(p) = p$. This is a diagnostic relationship, not a hypothesis test. To expose the sampling uncertainty caused by having only M assessed locations, let z be the $(1+\gamma)/2$ standard-normal quantile and set

$$d = 1 + z^2/M, \qquad a = \frac{\hat{c} + z^2/(2M)}{d}, \qquad h = \frac{z}{d}\left\{\frac{\hat{c}(1-\hat{c})}{M} + \frac{z^2}{4M^2}\right\}^{1/2}. \tag{V-16}$$

The reported Wilson interval is

$$[\max(0, a-h), \min(1, a+h)]. \tag{V-17}$$

These are pointwise finite-location bounds. They are not simultaneous bands, and spatial dependence between locations can reduce their effective information, so the bounds must not be read as a spatially adjusted confidence statement.

Display algorithms

The validation and simulation plot methods are deliberately views over stored, unrounded results. Cross-validation displays observed against predicted values, error against prediction, or error by fold. Predictive assessment displays the stored mean and interval against truth, or error against truth. Parameter recovery displays one named parameter across replicates, its stored truth and, when requested, Wald intervals formed from the already reported standard errors and recovery level. Plotting does not refit a model or alter any metric.

Code and validation map

Equations	Implementation	Independent checks
V-01	`sp_cross_validate()`	One fold reproduced by a separate manual refit; supplied fold IDs preserved
V-02–V-08	`sp_cross_validate()`, `sp_predictive_log_score()`	Gaussian direct density; independent binomial/Poisson Gaussian-link integration; finite fold scores
V-09–V-11	`sp_simulation_assess()`	Direct draw means, zero-RMSE fixture and interval coverage
V-12–V-13	`.sp_empirical_crps()`	Agreement with brute-force pairwise absolute differences

Equations	Implementation	Independent checks
V-14	`sp_simulation_assess()`	Direct exceedance frequencies
V-15–V-17	`sp_calibration()`	Direct type-8 interval coverage and independent Wilson-bound reconstruction
Display algorithms	validation and simulation `plot()` methods	Every supported view renders from fixed synthetic stored values

Buffered, regular-block and environmental fold constructors, Cartesian parameter-grid generation, scenario-level parallel recovery and structured reporting are now implemented and derived in the later chapters. Remaining work includes further domain-specific fold constructors only when a consumer use case requires them, and consumer-specific integration where authorized.

Monte Carlo maximum likelihood for non-Gaussian spatial models

Purpose and prerequisites

The Laplace fit replaces a difficult integral by a local quadratic approximation. This chapter derives the optional Monte Carlo maximum-likelihood (MCML) correction implemented by `sp_mcml()`. It estimates the same marginal likelihood integral directly by normalized importance sampling. The method is independently implemented from the equations below.

The reader should know probability densities, the multivariate normal distribution, logarithms, sample means and elementary maximum likelihood. The latent model and Laplace mode are derived in Chapters 4 and 5.

The marginal likelihood problem

Let $y = (y_1, \dots, y_n)^\mathsf{T}$, let s be the latent Gaussian vector, and write

$$s \sim N(0, \Sigma_\psi), \qquad \eta_i = x_i^\mathsf{T}\beta + o_i + s_i. \tag{C-01}$$

Here ψ contains spatial variance, scale and independent latent variance, while Matérn smoothness remains fixed. Conditional on s, observations are independent binomial-logit or Poisson-log. For $\theta = (\beta, \psi)$, define the joint density

$$h_\theta(s) = \left\{ \prod_{i=1}^{n} p(y_i \mid s_i, \beta) \right\} \phi_n(s; 0, \Sigma_\psi). \tag{C-02}$$

The marginal likelihood is the integral

$$L(\theta; y) = \int_{\mathbb{R}^n} h_\theta(s)\, ds. \tag{C-03}$$

This integral usually has no closed form. Laplace fitting approximates it near one mode. MCML instead evaluates it as an expectation.

A normalized importance distribution

At the supplied preliminary fit, let $\hat{s}_0$ be the latent posterior mode and let C_0 be the inverse negative posterior Hessian. For a positive proposal-scale constant a, SPmodels uses the fully normalized density

$$q(s) = \phi_n(s; \hat{s}_0, a^2 C_0). \tag{C-04}$$

This is a proposal distribution, not a claim that the posterior is exactly Gaussian. If $R^\mathsf{T} R = a^2 C_0$ and $z_b \sim N(0, I_n)$, independent proposal draws are

$$s_b = \hat{s}_0 + R^\mathsf{T} z_b, \qquad b = 1, \ldots, B. \tag{C-05}$$

One scale can be too narrow for the posterior tails or unnecessarily wide near its centre. With declared distinct scales $a_1, \ldots, a_K$, SPmodels therefore also supports the normalized mixture

$$q_{\text{mix}}(s) = \sum_{k=1}^{K} \alpha_k \phi_n(s; \hat{s}_0, a_k^2 C_0), \qquad \sum_k \alpha_k = 1. \tag{C-05a}$$

It assigns B_k draws deterministically to component k, balanced so the counts differ by at most one, and sets $\alpha_k = B_k/B$. Every draw is weighted by the complete mixture density in (C-05a), not merely by the density of the component that generated it. The deterministic-mixture estimator is

$$\widehat{L}_B = \sum_{k=1}^{K} \alpha_k \frac{1}{B_k} \sum_{b \in k} \frac{h_\theta(s_b)}{q_{\text{mix}}(s_b)}. \tag{C-05b}$$

Because $\alpha_k = B_k/B$, (C-05b) reduces algebraically to the ordinary mean in (C-08). Evaluation uses row-wise log-sum-exp over the component densities. This preserves normalization and avoids tail-density underflow.

Optional discarded-pilot adaptation

A local Gaussian approximation can have a useful shape but a poor centre or dispersion. When `adapt = TRUE`, SPmodels first draws an independent pilot sample $S_1^{(0)}, \ldots, S_{B_0}^{(0)}$ from the original normalized proposal q_0. At the preliminary parameter vector $\hat{\theta}_0$, it calculates

$$u_b = \frac{h_{\hat{\theta}_0}(S_b^{(0)})}{q_0(S_b^{(0)})}, \qquad \tilde{u}_b = \frac{u_b}{\sum_{c=1}^{B_0} u_c}. \tag{C-05c}$$

The importance-weighted pilot moments are

$$\bar{s}_w = \sum_{b=1}^{B_0} \tilde{u}_b S_b^{(0)}, \tag{C-05d}$$

$$C_w = \frac{\sum_{b=1}^{B_0} \tilde{u}_b (S_b^{(0)} - \bar{s}_w)(S_b^{(0)} - \bar{s}_w)^\mathsf{T}}{1 - \sum_{b=1}^{B_0} \tilde{u}_b^2}. \tag{C-05e}$$

The denominator is the usual normalized-weight correction. It approaches zero when one pilot draw carries almost all the weight, which is why the pilot ESS is reported and the calculation fails

explicitly if the covariance cannot be identified. For a declared shrinkage weight $0 < \lambda \leq 1$, the adapted base moments are

$$m_1 = \lambda \hat{s}_0 + (1-\lambda)\bar{s}_w, \qquad C_1 = \lambda C_0 + (1-\lambda)C_w. \tag{C-05f}$$

Positive λ regularizes the empirical covariance toward the positive- definite Laplace covariance. The retained proposal replaces the centre and base covariance in (C-05a) by m_1 and C_1, while preserving the declared scale factors and complete mixture density.

The pilot draws are not reused in the likelihood estimate. Conditional on the pilot, SPmodels draws a new independent main sample from the resulting normalized density q_1. Consequently

$$E\left\{ \frac{h_\theta(S)}{q_1(S)} \middle| \text{pilot} \right\} = \int h_\theta(s)\, ds = L(\theta; y). \tag{C-05g}$$

Thus adaptation changes the conditional variance, not the target integral. It is opt-in because noisy pilot weights can make moment estimates worse. If the pilot ESS fraction is below the declared `ess_warning` threshold, SPmodels warns and retains the original proposal moments; lowering that threshold is an explicit decision to trust a weaker pilot. The used/fallback decision and its reason, original and adapted moments, pilot draws, pilot weights and pilot diagnostics are retained so the decision can be audited.

Because q integrates to one and is positive wherever the Gaussian prior is positive, multiply and divide (C-03) by q:

$$L(\theta; y) = \int \frac{h_\theta(s)}{q(s)} q(s)\, ds = E_q\{w_\theta(S)\}, \tag{C-06}$$

where

$$w_\theta(s) = \frac{h_\theta(s)}{q(s)}. \tag{C-07}$$

Therefore the Monte Carlo likelihood estimator is

$$\widehat{L}_B(\theta) = \frac{1}{B} \sum_{b=1}^{B} w_\theta(s_b). \tag{C-08}$$

No unknown normalizing constant has been discarded. Consequently `logLik()` can report $\log \widehat{L}_B$, rather than only a relative likelihood.

The implemented log weight

For each proposal draw, SPmodels evaluates

$$\begin{aligned} \log w_\theta(s_b) = & \sum_{i=1}^{n} \log p(y_i \mid s_{bi}, \beta) \\ & - \frac{1}{2}\left\{ n \log(2\pi) + \log|\Sigma_\psi| + s_b^\top \Sigma_\psi^{-1} s_b \right\} \\ & - \log q(s_b). \end{aligned} \tag{C-09}$$

Cholesky solves compute the quadratic form; SPmodels does not explicitly invert the covariance for every draw. If $m = \max_b \log w_b$, the stable log-likelihood calculation is

$$\log \widehat{L}_B = m + \log \left\{ \frac{1}{B} \sum_{b=1}^{B} \exp(\log w_b - m) \right\}. \tag{C-10}$$

The same standard-normal draws are retained throughout one optimization. This common-random-number rule makes the approximated objective a deterministic function of θ during that call. Regression parameters are optimized on their ordinary scale. Positive covariance parameters are optimized on the log scale with the same finite numerical bounds as the preliminary fitter.

If a trial covariance cannot be factorized or its fixed-draw log weights are not finite, the minimized objective returns the finite sentinel 10^{20}. This prevents numerical finite-difference overflow at an invalid boundary; the final parameter vector must still yield a usable latent state, and the optimizer convergence code and importance diagnostics remain visible.

Importance-sampling diagnostics

Let normalized weights be

$$\widetilde{w}_b = \frac{w_b}{\sum_{c=1}^{B} w_c}.$$

The importance effective sample size and fraction are

$$B_{\text{eff}} = \frac{1}{\sum_b \widetilde{w}_b^2}, \qquad f_{\text{eff}} = \frac{B_{\text{eff}}}{B}. \tag{C-11}$$

They range from 1 to B, and from $1/B$ to 1, respectively. SPmodels also records $\max_b \widetilde{w}_b$. A small effective fraction or one dominant weight means that too few draws determine the answer.

For independent raw weights $w_1, \ldots, w_B$, let $\bar{w}$ and s_w be their sample mean and standard deviation. The relative Monte Carlo standard error is

$$r_{\text{MC}} = \frac{s_w}{\sqrt{B}\,\bar{w}}. \tag{C-12}$$

The delta method gives the same first-order value as the Monte Carlo standard error of $\log \widehat{L}_B$. Rescaling every weight by a common constant does not change (C-11) or (C-12), so both are calculated safely after subtracting the largest log weight.

For deterministic mixture sampling, the likelihood mean remains the same but its variance estimate respects the proposal-component strata:

$$\widehat{\text{Var}}(\widehat{L}_B) = \sum_{k=1}^{K} \alpha_k^2 \frac{s_{w,k}^2}{B_k}. \tag{C-12a}$$

SPmodels uses the square root of (C-12a) divided by $\widehat{L}_B$ for the mixture relative MCSE. Treating all stratified weights as identically distributed would use the wrong variance formula.

These diagnostics detect many failures but cannot prove that the proposal has covered every important part of the integrand. Results should be repeated with larger B, a changed proposal scale, and a refitted proposal when scientific conclusions are sensitive.

Independent replication

A diagnostic calculated from one importance sample cannot show how much the reported optimum moves when that sample changes. Let $\hat{\theta}_r$ be the optimizer from independent replicate $r = 1, \ldots, R$, with the proposal rule and sample size held fixed. For parameter component j, SPmodels reports

$$\bar{\theta}_j = \frac{1}{R}\sum_{r=1}^{R}\hat{\theta}_{rj}, \qquad s_{\text{between},j} = \left\{\frac{1}{R-1}\sum_{r=1}^{R}(\hat{\theta}_{rj} - \bar{\theta}_j)^2\right\}^{1/2}. \tag{C-13}$$

It also retains each replicate's log likelihood, effective sample size, largest normalized weight and relative MCSE. This is a sensitivity analysis: the replicate mean is not substituted for an MCML fit, and $s_{\text{between},j}$ is not the sampling standard error of a parameter. `sp_mcml_replicates()` returns all fitted objects so no convergence failure or diagnostic is hidden by the summary.

When discarded-pilot adaptation is requested, the replicate diagnostics also carry pilot ESS fraction, maximum weight, relative MCSE and whether the pilot passed the gate in (C-05c)–(C-05f). Precision-escalation histories record the same fields at every attempt. A fallback is therefore visible in the compact history as well as in the retained fitted object.

Let $\widehat{V}_r$ be the fixed-objective Hessian covariance from replicate r. SPmodels additionally records

$$\widehat{V}_{\text{MC}} = \frac{1}{R-1}\sum_{r=1}^{R}(\hat{\theta}_r - \bar{\theta})(\hat{\theta}_r - \bar{\theta})^{\mathsf{T}}, \tag{C-13a}$$

$$\widehat{V}_{\text{sampling}} = \frac{1}{R}\sum_{r=1}^{R}\widehat{V}_r, \qquad \widehat{V}_{\text{total}} = \widehat{V}_{\text{sampling}} + \widehat{V}_{\text{MC}}. \tag{C-13b}$$

Diagonal square roots give sampling, Monte Carlo and combined standard errors on the optimizer's parameter scale. SPmodels also transforms every replicate to natural model parameters and applies the diagonal Jacobian containing 1 for mean coefficients, the parameter value for log-scale components, and $\alpha(1-\alpha)$ for a logit intervention impact. Equations C-13a–C-13b are then repeated on that interpretable scale. The Monte Carlo standard error of the replicate mean is smaller by $\sqrt{R}$, but the package does not silently substitute that mean for a fitted model. If any replicate Hessian covariance is unusable, sampling and total entries remain missing while empirical Monte Carlo covariance is retained.

Precision-directed sample-size escalation

For an initial sample size B_1, maximum $B_{\max}$, and growth factor $g > 1$, define

$$B_{k+1} = \min\{B_{\max}, \max(B_k + 1, \lceil gB_k \rceil)\}. \tag{C-14}$$

Attempt k meets the declared targets when

$$r_{\text{MC},k} \le r_\star \quad \text{and} \quad f_{\text{eff},k} \ge f_\star, \tag{C-15}$$

where $r_\star > 0$ and $0 < f_\star < 1$. `sp_mcml_precision()` stops at the first success. If no attempt succeeds, it evaluates $B_{\max}$, returns that fit, records that the targets were not met and warns. It never relaxes a target to manufacture success.

Every attempt uses a new independent standard-normal sample. If proposal refresh is requested, the preceding estimate supplies a newly calculated latent mode $\hat{s}_k$ and local covariance C_k, giving

$$q_{k+1}(s) = \phi_n(s; \hat{s}_k, a^2 C_k). \tag{C-16}$$

Without refresh, every attempt uses the original proposal. Refresh can improve local overlap, but neither policy proves that all important regions were visited. The returned history records B_k, seed, both sides of (C-15), largest weight, log likelihood and the stopping decision for every attempt.

Curvature and downstream prediction

The observed Hessian of the fixed-draw objective supplies a local parameter- covariance approximation when it is positive definite. It describes curvature of that particular Monte Carlo objective. It does not automatically add the simulation error in (C-12) to every parameter standard error. Independent replicates are required before the separate and combined covariance estimates in (C-13a)–(C-13b) are available.

After optimization, SPmodels recomputes the latent mode and local covariance at the MCML estimates. Existing point and joint prediction can therefore run at the corrected parameter estimates, but their conditional latent calculations still use a local Gaussian approximation. MCML correction of parameter estimation does not make downstream latent prediction exact.

Cross-validation of an MCML fit repeats both the preliminary Laplace fit and the MCML correction inside every training fold. Held-out observations never enter either stage.

Algorithm

1. Require a fitted binomial-logit or Poisson-log spatial model.
2. Construct (C-04), or the declared mixture (C-05a), from its latent mode and local covariance.
3. If adaptation is requested, draw the independent pilot and diagnose it. If its ESS threshold passes, calculate (C-05c)–(C-05f) and construct the normalized adapted proposal; otherwise retain the original moments and record the fallback.
4. Discard the pilot from likelihood estimation and generate the independent, retained component-balanced sample. Equation (C-05g) preserves the target.
5. For every trial parameter vector, construct the Matérn covariance and evaluate (C-09)–(C-10).
6. Maximize the fixed-draw log likelihood using bounded log covariance parameters.
7. Recalculate final weights, (C-11)–(C-12), Hessian information and the local latent state.
8. Warn, without silently changing the criterion, when the effective-sample fraction is below the declared threshold.
9. For independent replication, repeat the preceding steps with independent draws and calculate (C-13)–(C-13b).
10. For precision escalation, apply (C-14)–(C-16), retaining every attempt and stopping only at the declared targets or maximum.

Validation map

Equations	Implementation	Independent checks
C-01–C-03	existing non-Gaussian model state	Previously reconstructed likelihood and prior terms
C-04–C-10	`sp_mcml()` and `.sp_mcml_log_weights()`	One-dimensional binomial integration; direct normalized mixture density, balanced allocation and finite weights

Equations	Implementation	Independent checks
C-05c–C-05g	`.sp_mcml_adapt_proposal()` and `sp_mcml()`	Exact seeded pilot/main separation, weighted moments, positive-definite regularization, direct adapted-mixture density and full-shrinkage boundary
C-11–C-12a	`.sp_mcml_diagnostics()`	Bounds, stratified mixture error, direct likelihood reconstruction and exact seeded reproducibility
C-13–C-13b	`sp_mcml_replicates()`	Direct replicate means, MC covariance, mean MCSE and combined covariance; independent seeds and caller-seed preservation
C-14–C-16	`sp_mcml_precision()`	First-attempt success, exact capped escalation, history and invalid-control failures
Complete algorithm	`sp_mcml()`	Binomial and Poisson fits; local-seed preservation; downstream prediction; MCML repeated inside folds

Scope and references

The implementation is dense and intended for small to moderate point datasets. Each attempt may use one Gaussian proposal or a normalized same-centre mixture of declared scales. Independent replicates, proposal refresh between attempts, parameter-level Monte Carlo covariance and automatic sample-size escalation are available. Optional within-attempt adaptation uses an independently drawn, discarded pilot and regularized importance-weighted moments; it is not assumed to improve every proposal. Bridge sampling and large-data sparse matrix methods are not implemented.

The general Monte Carlo maximum-likelihood justification follows Geyer (1994), *Journal of the Royal Statistical Society: Series B*, 56(1), 261–274, doi:10.1111/j.2517-6161.1994.tb01976.x. Spatial generalized-linear-model use is discussed by Christensen (2004), *Journal of Computational and Graphical Statistics*, 13(3), 702–718, doi:10.1198/106186004X2525. These citations support the statistical method; the derivation, interface and implementation here are SPmodels' own.

Non-Gaussian simulation, residuals and spatial folds

Purpose and prerequisites

This chapter derives fitted response simulation and residual diagnostics for binomial-logit and Poisson-log spatial models, then specifies the coordinate clustering used to construct spatial cross-validation folds. The reader should know conditional probability, binomial and Poisson means and variances, and squared Euclidean distance.

Fitted-model response simulation

For fitted parameters $\widehat{\beta}$ and $\widehat{\psi}$, let

$$s^{(b)} \sim N(0, \Sigma_{\widehat{\psi}}), \qquad b = 1, \ldots, B. \tag{S-01}$$

If $R^{\mathsf{T}} R = \Sigma_{\widehat{\psi}}$, SPmodels generates

$$z^{(b)} \sim N(0, I), \qquad s^{(b)} = R^{\mathsf{T}} z^{(b)}. \tag{S-02}$$

The simulated linear predictor is

$$\eta_i^{(b)} = x_i^{\mathsf{T}} \widehat{\beta} + o_i + s_i^{(b)}. \tag{S-03}$$

For a binomial model with fitted trial total m_i, simulation continues as

$$p_i^{(b)} = \text{logit}^{-1}(\eta_i^{(b)}), \qquad Y_i^{(b)} \mid s_i^{(b)} \sim \text{Binomial}(m_i, p_i^{(b)}). \tag{S-04}$$

The returned value is a success count, including when the original response was entered as a proportion. For Poisson-log models,

$$\lambda_i^{(b)} = \exp(\eta_i^{(b)}), \qquad Y_i^{(b)} \mid s_i^{(b)} \sim \text{Poisson}(\lambda_i^{(b)}). \tag{S-05}$$

Equations (S-01)–(S-05) are unconditional on the observed responses and conditional on fitted parameters. They are appropriate for fitted-model checks and simulation studies. They do not draw parameter uncertainty. A supplied seed is scoped to the function and the caller's random-number state is restored.

Conditional residuals at the latent mode

Let $\widehat{\mu}_i$ be the fitted conditional response mean at the final latent mode. The raw response residual is

$$r_i = y_i - \widehat{\mu}_i. \tag{S-06}$$

The conditional response variance is

$$V_i = \begin{cases} m_i \widehat{p}_i (1 - \widehat{p}_i), & \text{binomial}, \\ \widehat{\mu}_i, & \text{Poisson}, \end{cases} \qquad \widehat{p}_i = \widehat{\mu}_i / m_i. \tag{S-07}$$

When $V_i > 0$, the Pearson residual is

$$r_i^P = \frac{y_i - \widehat{\mu}_i}{\sqrt{V_i}}. \tag{S-08}$$

Zero conditional variance has no finite Pearson standardization and is reported as missing rather than replaced by an arbitrary number.

For Poisson data, the deviance contribution is

$$d_i^2 = 2 \left[y_i \log \left(\frac{y_i}{\widehat{\mu}_i} \right) - (y_i - \widehat{\mu}_i) \right], \tag{S-09}$$

with $0 \log 0 = 0$. For binomial counts,

$$d_i^2 = 2\left[y_i \log\left(\frac{y_i}{\widehat{\mu}_i}\right) + (m_i - y_i)\log\left(\frac{m_i - y_i}{m_i - \widehat{\mu}_i}\right)\right], \tag{S-10}$$

with the analogous zero-count conventions. The signed deviance residual is

$$r_i^D = \operatorname{sign}(r_i)\sqrt{d_i^2}. \tag{S-11}$$

For a discrete fitted conditional CDF F_i, define

$$a_i = F_i(y_i - 1), \qquad b_i = F_i(y_i). \tag{S-11a}$$

Draw a caller-seeded value and transform it:

$$U_i \sim \text{Uniform}(a_i, b_i), \qquad r_i^Q = \Phi^{-1}(U_i). \tag{S-11b}$$

Under the correct discrete conditional distribution, the probability intervals partition $[0, 1]$. Mixing over them makes U_i marginally uniform, hence

$$r_i^Q \sim N(0, 1). \tag{S-11c}$$

Probabilities are clipped only at machine epsilon before Φ^{-1}, preventing infinite computer values. These residuals remain conditional on the fitted latent mode and parameters; spatial dependence and parameter fitting are not removed. They are distinct from Gaussian Cholesky-decorrelated residuals.

For any chosen non-Gaussian residual r_i above, the empirical residual variogram in distance bin P_k is

$$\hat{\gamma}_k^{(r)} = \frac{1}{|P_k|}\sum_{(i,j)\in P_k} \frac{1}{2}(r_i - r_j)^2. \tag{S-11d}$$

SPmodels supports Pearson, signed deviance and caller-seeded randomized- quantile residuals. It returns pair counts and the empirical curve but no fitted response-scale curve: the latent link covariance is not silently treated as a semivariogram for discrete residuals.

For a permutation envelope, replicate b applies a uniformly sampled permutation π_b to the selected residual vector and recomputes (S-11d):

$$\hat{\gamma}_{kb}^{\text{perm}} = \frac{1}{|P_k|}\sum_{(i,j)\in P_k} \frac{1}{2}(r_{\pi_b(i)} - r_{\pi_b(j)})^2. \tag{S-11e}$$

Pointwise type-8 quantiles, the standardized global curve statistic and the finite Monte Carlo p-value then follow equations (D-07)–(D-10). The null is exchangeability of the chosen residual values across fixed locations. This is a conditional diagnostic, not a proof of model adequacy; heterogeneous or poorly calibrated residual distributions weaken that interpretation.

For a fitted-model parametric envelope, replicate b instead follows the complete discrete-data workflow

$$Y_b^* \sim p(\cdot \mid \widehat{\theta}), \qquad \widehat{\theta}_b^* = \text{fit}(Y_b^*), \qquad \hat{\gamma}_{kb}^{\text{par}} = \frac{1}{|P_k|}\sum_{(i,j)\in P_k} \frac{1}{2}(r_{ib}^* - r_{jb}^*)^2. \tag{S-11f}$$

Here `fit` repeats the model's declared Laplace or Monte Carlo ML route, and r^*_{ib} is the chosen Pearson, deviance or randomized-quantile residual from that refit. Thus the envelope includes response simulation and refitting variation rather than reusing the original latent mode. Every optimizer convergence code and captured warning is retained. A non-zero code is reported to the caller and must be examined; SPmodels does not relabel it as success. The pointwise and global summaries again use (D-07)–(D-10).

Geometry-driven spatial folds

Random folds can place nearby or repeated locations on both sides of a validation split. To prevent exact-location leakage, define coordinate equivalence classes

$$i \sim j \quad \Longleftrightarrow \quad c_i = c_j. \tag{S-12}$$

SPmodels clusters the U unique coordinate vectors, not all n rows. For requested fold count $K \leq U$, k-means selects clusters $C_1, \ldots, C_K$ and centroids $a_1, \ldots, a_K$ to reduce

$$\sum_{k=1}^{K} \sum_{u \in C_k} \|c_u - a_k\|^2. \tag{S-13}$$

All original rows in one equivalence class inherit the same cluster. Cluster numbers from k-means are arbitrary, so SPmodels canonicalizes labels by sorting centroids from largest to smallest second coordinate, then smallest to largest first coordinate. This makes labels interpretable as top-to-bottom and left-to-right; the partition itself can still depend on initialization.

Regular spatial blocks

For declared positive axis widths w_1, w_2 and grid origin $o = (o_1, o_2)$, row i belongs to occupied block

$$B_i = \left(\left\lfloor \frac{c_{i1} - o_1}{w_1} \right\rfloor, \left\lfloor \frac{c_{i2} - o_2}{w_2} \right\rfloor \right). \tag{S-13c}$$

Every row with the same B_i, including repeated coordinates, receives one fold label. Occupied blocks are ordered top-to-bottom then left-to-right and assigned round-robin, or the occupied-block order is locally seeded and permuted before the same balanced allocation:

$$f_i = a(B_i), \qquad a(B) \in \{1, \ldots, K\}. \tag{S-13d}$$

The operation balances numbers of occupied blocks, not necessarily numbers of observations or outcomes. It retains block indices, row counts, widths, origin, units and allocation rule so the split is auditable. Blocks control which supports are withheld together; they do not impose a distance gap.

Environmental-space folds

Let e_{ij} be environmental variable j at unique location i. With the default standardization, clustering uses

$$z_{ij} = \frac{e_{ij} - \bar{e}_j}{s_j}, \tag{S-13e}$$

where a zero-variance variable is rejected. K-means then chooses environmental clusters $G_1, \dots, G_K$ and centres m_k to reduce

$$\sum_{k=1}^{K} \sum_{i \in G_k} \|z_i - m_k\|^2. \tag{S-13f}$$

Rows at an identical coordinate must have identical declared environmental values and inherit one cluster, preventing exact-location leakage. Centres are canonically ordered lexicographically, and the seed is local. These folds test transfer across environmental regimes; they need not be geographically compact and do not impose a distance buffer.

Group-constrained folds

Let $g(i) \in \{1, \dots, G\}$ identify the declared clinic, survey, school or other grouping unit for row i, and let

$$n_g = |\{i : g(i) = g\}|. \tag{S-13g}$$

Every group is indivisible. In deterministic mode, groups are considered from largest to smallest n_g, with first appearance breaking ties. In random mode their order is locally seeded and permuted. If L_k is the number of rows already assigned to fold k, the next group goes to

$$k^* = \arg\min_{1 \le k \le K} L_k, \qquad L_{k^*} \leftarrow L_{k^*} + n_g. \tag{S-13h}$$

The first fold breaks load ties. This greedy rule targets row balance without ever splitting a group; exact equality is not promised when group sizes differ. Returned metadata records every group label, row count and allocation.

Buffered training sets

For test fold $\mathcal{H}_k$ and positive buffer distance b, define the fold-specific training set

$$\mathcal{T}_k(b) = \left\{ i : f_i \neq k, \ \min_{j \in \mathcal{H}_k} d_{ij} > b \right\}. \tag{S-13a}$$

Rows with $f_i \neq k$ that fall inside the buffer are neither used for training nor scored as test observations in fold k. The held-out set stays $\mathcal{H}_k$. The exclusion set is therefore

$$\mathcal{E}_k(b) = \{i : f_i \neq k\} \setminus \mathcal{T}_k(b). \tag{S-13b}$$

The buffer uses the fitted coordinate units and is applied from every test location. This reduces short-range leakage but can greatly reduce the training sample. A fold with no remaining training rows fails explicitly.

Multiple random starts reduce sensitivity to poor local k-means solutions. A supplied seed is local. These folds are spatially compact but are not promised to have equal observation counts, equal response prevalence or a minimum between-fold separation. Scientific applications should inspect the returned centres and fold sizes.

Repeated simulation and parameter recovery

The same fitted-model generator supports a repeated recovery experiment. Let $\hat{\theta}$ be the supplied fitted parameter vector. For replicate b, generate

$$Y^{(b)} \sim p(y \mid \hat{\theta}) \tag{S-14}$$

using the family-specific hierarchy above, then refit the same declared model to obtain $\hat{\theta}^{(b)}$. If the supplied fit used MCML, both the preliminary Laplace stage and MCML correction are repeated.

The recovery vector contains every parameter estimated by the supplied route: mean coefficients; spatial and nugget covariance parameters; grouped random-intercept variances; temporal scale; and intervention impact/decay when their corresponding estimation flags are true. Fixed quantities are retained by each refit but are not mislabelled as recovered estimates. Qualified `random::` names prevent a group variance from colliding with a mean or spatial parameter name. For parameter j, define replicate error

$$e_{bj} = \hat{\theta}_j^{(b)} - \hat{\theta}_j. \tag{S-15}$$

Across B_j successful refits, estimated bias and RMSE are

$$\widehat{\mathrm{Bias}}_j = \frac{1}{B_j}\sum_b e_{bj}, \qquad \widehat{\mathrm{RMSE}}_j = \left(\frac{1}{B_j}\sum_b e_{bj}^2\right)^{1/2}. \tag{S-16}$$

The empirical standard deviation is

$$s_j = \left\{\frac{1}{B_j - 1}\sum_b (\hat{\theta}_j^{(b)} - \bar{\hat{\theta}}_j)^2\right\}^{1/2}. \tag{S-17}$$

If replicate b reports usable standard error $\widehat{\mathrm{se}}_{bj}$, SPmodels also calculates their mean and the Wald coverage indicator

$$I_{bj} = \mathbb{1}\left[\hat{\theta}_j^{(b)} - z_{(1+\gamma)/2}\widehat{\mathrm{se}}_{bj} \le \hat{\theta}_j \le \hat{\theta}_j^{(b)} + z_{(1+\gamma)/2}\widehat{\mathrm{se}}_{bj}\right], \tag{S-18}$$

and coverage

$$\widehat{C}_j = \frac{1}{B_j^{\mathrm{se}}}\sum_b I_{bj}. \tag{S-19}$$

Failed refits are never converted to successful values. They are recorded with their messages, and every parameter summary reports its successful count. The caller may instead request immediate failure.

The response matrix is generated before any optional forked dispatch. For replicate b, MCML receives the deterministic seed

$$s_b = (s + b) \bmod M, \tag{S-19a}$$

where s is the declared base seed and M is R's maximum integer. Therefore worker scheduling cannot change simulated responses or stochastic refit streams, and sequential and parallel estimates

agree exactly. Parallel MCML requires an explicit base seed; Windows currently uses one worker. The caller's random-number state is restored before workers are launched.

This experiment treats the fitted values as truth. It measures recovery near one fitted scenario, conditional on its design, coordinates and offsets. It is not a substitute for a predeclared simulation grid spanning scientifically important parameter values.

Multiple fitted scenarios

Let $q = 1, \ldots, Q$ index named fitted models. Each model defines its own truth $\hat{\theta}_q$, design, covariance, family and replicate count B_q. SPmodels runs the preceding experiment separately:

$$\mathcal{R}_q = \text{Recovery}(\hat{\theta}_q, B_q, s_q), \qquad s_q = s_1 + q - 1, \tag{S-20}$$

where integer seeds wrap at R's maximum integer. There is no pooling across scenarios: rows are stacked only after attaching the scenario identifier. Every `spmodels_simulation_recovery` object, simulated response, failure and estimate remains stored. This orchestration makes a predeclared simulation design easier to audit, but does not make an arbitrary list of scenarios scientifically comprehensive. Scenario seeds are resolved before dispatch. Consequently, the sequential route and optional non-Windows forked scenario workers produce identical per-scenario experiments; parallel scheduling does not choose the random streams.

Precision-directed replicate escalation

For attempt a, let B_a be its declared replicate count and let B_{aj} estimates of parameter j be finite. The Monte Carlo standard error of estimated mean bias is estimated by

$$\widehat{\text{MCSE}}_{aj} = \frac{s_{aj}}{\sqrt{B_{aj}}}, \tag{S-21}$$

where s_{aj} is the empirical standard deviation in (S-17). An attempt passes only if every recovered parameter has at least the declared minimum successful count and

$$\widehat{\text{MCSE}}_{aj} \leq \epsilon \quad \text{for every } j. \tag{S-22}$$

Otherwise the requested count grows geometrically and is capped exactly at $B_{\max}$. Every attempt is a complete independently seeded recovery experiment; its count, seed, minimum successes, worst finite MCSE and status are retained. A failed capped attempt remains a failure. The threshold is never relaxed to obtain a pass. Each complete attempt may use the replicate workers in (S-19a); tests require its estimates and stopping history to equal the sequential route exactly.

Cartesian parameter scenarios

Let parameter j have a finite declared set of truths

$$\Theta_j = \{\theta_{j1}, \ldots, \theta_{jm_j}\}. \tag{S-23}$$

`sp_simulation_grid()` constructs the Cartesian product

$$\mathcal{G} = \Theta_1 \times \cdots \times \Theta_q, \qquad |\mathcal{G}| = \prod_{j=1}^{q} m_j. \tag{S-24}$$

R's first grid column varies fastest, and SPmodels records the resulting table as an attribute in exactly the returned scenario order. Only parameters that the unchanged refit specification can recover are accepted. This prevents a misleading experiment in which data are generated under one fixed covariance but every replicate is refitted under a different fixed covariance.

For each row of $\mathcal{G}$, SPmodels replaces the declared coefficient, covariance, random-variance, temporal or intervention values and rebuilds the generative covariance factor. For Gaussian models this is the Cholesky factor of V in (S-01). For binomial and Poisson models it is the Cholesky factor of Σ, followed by recomputation of the latent mode so the scenario is internally coherent. The original formula, locations, response family, trial totals and fitting controls are retained. These objects define simulation truths; they are not presented as new fits to the observed response.

Code and validation map

Equations	Implementation	Independent checks
S-01–S-05	`simulate.spmodels_fit()`	Exact reconstruction from seeded normal and response-family RNG draws; response support
S-06–S-11c	`sp_diagnostics()`	Direct mean, variance, Pearson and deviance calculations; exact seeded CDF-interval uniform and normal transform
S-11d	`sp_variogram()`	Direct binning of seeded randomized-quantile pair semivariances, pair counts, RNG isolation and rendered empirical-only display
S-11e–S-11f and D-07–D-10	`sp_variogram_envelope()`	Exact reconstruction of every seeded non-Gaussian permutation curve and the first full simulate-refit parametric curve, deterministic repeats, retained refit convergence/warnings, finite p-value bounds and empirical-only display
S-12–S-13	`sp_spatial_folds()`	Duplicate coordinates remain together, local seeded reproducibility, canonical labels and cross-validation use
S-13c–S-13d	`sp_block_folds()`, `sp_cross_validate()`	Direct block-index reconstruction, whole-block retention, deterministic/random allocation, RNG isolation and cross-validation use
S-13e–S-13f	`sp_environmental_folds()`, `sp_cross_validate()`	Formula/name equivalence, standardized centres, local RNG isolation and environmental-fold cross-validation use

Equations	Implementation	Independent checks
S-13g–S-13h	`sp_group_folds()`, `sp_cross_validate()`	Direct greedy allocation, whole-group retention, local seeded reproducibility, retained metadata and complete cross-validation use
S-13a–S-13b	`sp_buffered_folds()`, `sp_cross_validate()`	Every near-test row excluded; one fold reproduced by a direct manually buffered refit; oversized buffer failure
S-14–S-19a	`sp_simulation_recovery()`	First replicate matched to an independent refit; direct bias/RMSE reconstruction; binomial count/trial support; exact sequential/parallel result equality and RNG isolation
S-20	`sp_simulation_scenarios()`	Each sequential or parallel scenario exactly matches its direct seeded recovery call; input order, unequal replicate counts, RNG isolation and combined labels
S-21–S-22	`sp_simulation_precision()`	Direct MCSE reconstruction, first-attempt pass, geometric escalation, exact cap, RNG isolation and exact sequential/parallel attempt equality
S-23–S-24	`sp_simulation_grid()`	Exact Cartesian order, parameter substitution, direct Gaussian covariance reconstruction, non-Gaussian Cholesky reconstruction, response support and rejection of fixed/non-recoverable parameters

Limitations

Simulation is dense and uses the fitted observation locations and trial totals. It supports declared new parameter values over Cartesian grids but does not yet simulate new spatial supports. Non-Gaussian diagnostics include empirical residual variograms and permutation or full simulate-refit parametric envelopes/global curve tests. Spatial fold test groups can use coordinate k-means, regular axis-aligned blocks or environmental-space clusters; group-constrained allocation is also available, while arbitrary user-supplied fold identifiers remain accepted. Further domain-specific constructors should be added only for a declared use case. Buffered training exclusion is implemented around geometry-driven k-means test groups. Repeated recovery supports named response variables, two-column binomial responses, Cartesian parameter grids and multiple named fitted scenarios with unequal replicate counts. Precision-directed replicate escalation is available for one fitted scenario. Named scenarios and individual replicates can run in parallel on non-Windows systems with preassigned seeds.

Structured reporting and consumer contracts

Purpose

Reporting functions do not introduce a new fitted probability model. They provide stable transformations of already defined SPmodels results so that an application, lesson or report need not inspect internal object fields.

Model table

For mean parameter $\hat{\beta}_j$ with reported covariance matrix $V_{\hat{\beta}}$, the table records

$$\mathrm{SE}(\hat{\beta}_j) = \sqrt{(V_{\hat{\beta}})_{jj}}, \qquad z_j = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)}, \qquad p_j = 2\Phi(-|z_j|).$$

These are large-sample Wald summaries, not exact finite-sample tests. Covariance parameters retain their fitted standard errors but receive no automatic Wald p-value because boundary constraints make that interpretation unreliable. Every row states whether a value was estimated or fixed. The Matérn scale row also carries the fitted coordinate distance unit.

Implementation: `R/reporting.R`, function `sp_model_table()`.

Prediction table

For stored joint draws $t_j^{(1)}, \ldots, t_j^{(B)}$ at target j, the reported quantities are

$$\bar{t}_j = B^{-1} \sum_b t_j^{(b)}, \qquad s_j = \left\{ (B-1)^{-1} \sum_b (t_j^{(b)} - \bar{t}_j)^2 \right\}^{1/2},$$

the type-8 empirical quantiles used throughout SPmodels, and, for threshold c,

$$\widehat{P}(T_j > c) = B^{-1} \sum_b I(t_j^{(b)} > c).$$

Grouped targets are first calculated inside every joint draw using the weights defined in derivation 06. This preserves spatial dependence and is different from aggregating marginal means after prediction.

Implementation: `R/reporting.R`, function `sp_prediction_table()`, delegating the calculations to `sp_predict_target()`.

Validation table

`sp_validation_table()` changes representation only. It reshapes previously calculated validation metrics into the common tuple

$$(\text{assessment}, \text{scope}, \text{id}, \text{metric}, \text{value}).$$

The definitions of RMSE, MAE, log score, coverage, CRPS, recovery bias and recovery RMSE remain those in derivations 07 and 09. No rounding occurs in the machine-readable table.

Presentation formatting

`sp_format_table()` first obtains one of the three stable tables above, or accepts an already constructed data frame. Let v be a finite numerical value and d the requested number of decimal places. Its displayed value is the fixed-decimal representation

$$F_d(v) = \text{format}(v, \text{fixed}, d). \tag{R-01}$$

For a p-value column, values below 10^{-d_p} are displayed as `<0.00...1` at precision d_p, rather than as zero. Missing numerical and character values use the declared missing-value marker. This operation changes representation only: the original unrounded data frame is retained verbatim in the `raw_table` attribute of the formatted view.

Assembled report

`sp_report()` constructs its statistical sections only by calling `sp_model_table()`, `sp_prediction_table()` and `sp_validation_table()`. The result stores two parallel collections:

1. the unrounded, machine-readable source tables; and
2. formatted character views produced from those same tables.

The report printer adds a title, basic analysis metadata and section labels. It does not refit, predict, resample, recalculate metrics or replace raw numbers with rounded ones. The resulting object can therefore support a readable console report while downstream applications continue to consume exact values.

Code and validation map

Operation	Implementation	Independent checks
Model estimates	`sp_model_table()`	Coefficients, covariance values, units and status reconstructed directly
Prediction summaries	`sp_prediction_table()`	Draw means, SD, quantiles and exceedance frequencies reconstructed directly
Validation schema	`sp_validation_table()`	Overall, fold and parameter scopes checked against source summaries
R-01 and p-value threshold	`sp_format_table()`	Exact fixed-decimal strings, threshold label and missing marker
Report assembly	`sp_report()`	Every raw section is identical to its stable public table; printed sections are present

Tests and limitations

`tests/test-reporting.R` checks values against direct calculations, validates the fixed schemas and exercises Gaussian prediction and cross-validation. `tests/test-report-presentation.R` ad-

ditionally checks formatting and assembled reports. The current report is a dependency-free console/report object, not a typeset journal template, HTML site or office-document exporter. Formatting remains a view over the unrounded tables.

Spatial audit utilities, variogram envelopes and area targets

Purpose and prerequisites

This chapter derives and specifies four supporting operations: auditable distance summaries, projected UTM selection, Monte Carlo variogram envelopes, and exact fractional-area aggregation. These operations do not introduce a new response model. They make spatial inputs and model checks explicit enough for research reports and MSc teaching.

The reader should know Euclidean distance, empirical quantiles, coordinate reference systems, Monte Carlo simulation and elementary polygon area.

Pairwise distance summary

For two-dimensional projected coordinates $s_i = (s_{i1}, s_{i2})$, the unordered-pair distance is

$$d_{ij} = \sqrt{(s_{i1} - s_{j1})^2 + (s_{i2} - s_{j2})^2}, \qquad 1 \le j < i \le n. \tag{D-01}$$

There are $n(n-1)/2$ pairs. A repeated location is defined by exact equality of both supplied coordinate values, not by an undocumented proximity tolerance. Its distance is zero. The caller must choose one of three policies:

1. include zero-distance pairs in every summary;
2. exclude them while retaining their count; or
3. stop with an error.

The returned mean and sample standard deviation use the retained distances. Quantiles use R's type-8 definition. If $p \in [0, 1]$, set

$$h = (m + 1/3)p + 1/3, \tag{D-02}$$

where m is the retained pair count, and interpolate between the adjacent ordered distances at position h, with the usual endpoint truncation. The coordinate unit is never guessed for an ordinary data frame; it is inherited only from a fitted model or projected `sf` CRS.

Implementation: `sp_distance_summary()` in `R/geometry-utils.R`.

UTM coordinate-reference selection

Ordinary zone rule

After transforming all geometry vertices to WGS84 longitude/latitude, the ordinary six-degree zone for longitude λ is

$$z(\lambda) = \min\{60, \max[1, \lfloor(\lambda + 180)/6\rfloor + 1]\}. \tag{D-03}$$

Thus an exact interior six-degree boundary belongs to the eastern zone and longitude 180 degrees is capped at zone 60.

Standard high-latitude exceptions

The implemented rule assigns zone 32 for 56–64 degrees north and 3–12 degrees east. From 72–84 degrees north, it assigns zones 31, 33, 35 and 37 to longitude bands 0–9, 9–21, 21–33 and 33–42 degrees east respectively. These rules prevent the ordinary formula from selecting the narrow zones that are replaced in Norway and Svalbard.

UTM selection is rejected south of 80 degrees south or north of 84 degrees north. A longitude span above 180 degrees is treated as an antimeridian crossing and rejected because an arithmetic bounding-box centre is then misleading.

Every coordinate is assigned a zone. Multiple observed zones cause an error by default. With the explicit centre policy, the zone is instead evaluated at

$$\lambda_c = (\lambda_{\min} + \lambda_{\max})/2, \qquad \varphi_c = (\varphi_{\min} + \varphi_{\max})/2. \tag{D-04}$$

Likewise, a geometry spanning the equator errors by default. The explicit centre policy uses the sign of φ_c, with zero assigned north. The WGS84 EPSG code is

$$\text{EPSG} = \begin{cases} 32600 + z, & \text{northern hemisphere,} \\ 32700 + z, & \text{southern hemisphere.} \end{cases} \tag{D-05}$$

The result records all observed zones, centre, policies and equator status so the choice remains auditable. It selects a CRS; the user still performs the visible transformation with `sf::st_transform()`.

Implementation: `sp_utm_crs()` in `R/geometry-utils.R`.

Residual-variogram Monte Carlo envelope

Let $\hat{\gamma}_k$ be the observed mean semivariance in distance bin k, defined in derivation 03. SPmodels supports two null-generating rules.

Permutation null

For the no-location-association null, keep coordinates and distance bins fixed and randomly permute the selected residual vector:

$$e^{(b)} = \pi_b(e), \qquad b = 1, \ldots, B. \tag{D-06}$$

This preserves the exact observed residual distribution but breaks its association with location. It addresses whether residual spatial arrangement is more structured than expected under exchangeability.

Fitted-model null

The parametric rule draws

$$Y^{(b)} \sim N(X\hat{\beta} + o, \widehat{V}) \tag{D-07}$$

with fitted parameters held fixed. Mean residuals subtract the fitted mean. Conditional residuals additionally subtract the fixed-parameter spatial conditional adjustment

$$\widehat{K}\widehat{V}^{-1}(Y^{(b)} - X\hat{\beta} - o). \tag{D-08}$$

This checks behaviour under the fitted model but does not include parameter re-estimation uncertainty.

For every simulation, use the same pairs and bins to obtain $\hat{\gamma}_k^{(b)}$. A pointwise level $1-\alpha$ envelope is

$$[q_{\alpha/2,k}, q_{1-\alpha/2,k}], \tag{D-09}$$

where the endpoints are type-8 empirical quantiles of the simulated bin values. Pointwise limits are not a simultaneous confidence band.

For a global curve comparison, let $\bar{\gamma}_k$ and s_k be the simulated mean and standard deviation in every populated, variable bin. Define

$$T_{\text{obs}} = \sum_k \left(\frac{\hat{\gamma}_k - \bar{\gamma}_k}{s_k} \right)^2, \qquad T_b = \sum_k \left(\frac{\hat{\gamma}_k^{(b)} - \bar{\gamma}_k}{s_k} \right)^2. \tag{D-10}$$

The finite-simulation p-value is

$$p = \frac{1 + \sum_{b=1}^{B} I(T_b \geq T_{\text{obs}})}{B+1}. \tag{D-11}$$

The added one prevents a reported p-value of zero. A seed is local to the call, and raw simulated curves are returned so every limit and statistic can be reconstructed.

Implementation: `sp_variogram_envelope()` in `R/assessment.R`.

Exact fractional-area polygon targets

Let prediction cell C_i and target region A_g be projected polygons. Their exact intersection area is

$$a_{ig} = |C_i \cap A_g|, \qquad f_{ig} = a_{ig}/|C_i|. \tag{D-12}$$

The fraction f_{ig} is retained for auditing. With optional non-negative cell density d_i, define raw mass weight

$$w_{ig} = a_{ig} d_i; \tag{D-13}$$

without a density, $d_i = 1$. As in derivation 06, weights are normalized within each region and the target is calculated inside every joint draw:

$$T_g^{(b)} = \frac{\sum_i w_{ig} Y_i^{(b)}}{\sum_i w_{ig}}. \tag{D-14}$$

A boundary-crossing cell can therefore contribute to several regions without duplicating its full weight. Target polygons must not overlap, all geometry must share one known projected CRS, and intersections with zero area are discarded. The caller explicitly chooses whether unmatched cells and declared regions with no positive-area intersection cause an error or are recorded and dropped. Dropping is never silent.

`sp_target_geometry()` then joins the already calculated, unrounded target table to point or polygon `sf` geometry by exact unique identifiers or an explicit row-order contract. Its plot method is only a view of those values.

Implementation: `sp_polygon_area_weights()` and `sp_target_geometry()` in `R/spatial-targets.R`.

Code and validation map

Equations	Implementation	Independent checks
D-01–D-02	`sp_distance_summary()`	Complete four-location distance enumeration and all duplicate policies
D-03–D-05	`sp_utm_crs()`	North/south EPSG codes, six-degree edge, longitude 180, Norway, Svalbard, multi-zone, equator, polar and antimeridian cases
D-06–D-11	`sp_variogram_envelope()`	Exact seeded reconstruction of every permutation curve, standardized statistics and p-value; parametric reproducibility
D-12–D-14	`sp_polygon_area_weights()`	Direct split-cell areas/fractions, density weighting, draw-level aggregation, unmatched cell and empty-region policies
Join contract	`sp_target_geometry()`	Reordered identifier match, attached numerical values and mismatch failure

The relevant tests are `tests/test-geometry-utils.R`, `tests/test-variogram-envelope.R` and `tests/test-spatial-targets.R`.

Limitations

UTM is a practical local projection choice, not a universal answer for a study spanning zones, the equator, polar areas or the antimeridian. Centre policies must be scientifically justified. Parametric variogram envelopes condition on fitted parameters. Fractional-area density weighting assumes the supplied density is constant within each prediction cell. The target plot uses base `sf` display and does not choose palettes or cartographic annotations.

Prediction scenario updates

Purpose and boundary

A prediction scenario asks what the fitted model predicts if selected covariate or exposure values are changed while the fitted parameters, observation data, locations and every unmentioned prediction-data value remain fixed. It is not a refit, a causal-effect estimator or an intervention

model. The calculation is useful for transparent sensitivity exercises and for constructing matched baseline/scenario maps.

`sp_prediction_update()` implements this contract. It accepts only variables used on the right-hand side of the fitted formula, including variables nested inside `offset()`. It rejects separately supplied fitting offsets because a safe prediction-data mapping for those values does not exist.

Row mapping and replacement

Let the prediction data contain rows $i = 1, \dots, m$, with unique stored row identifiers r_i. The caller selects a non-empty index set

$$\mathcal{J} \subseteq \{1, \dots, m\}. \tag{P-01}$$

Logical selection must have length m. Integer selection must contain valid, unique indices. Character selection is matched exactly to the row identifiers; partial matching is never used. If replacement values are named, their names must be exactly the selected identifier set and the values are reordered to the selected data rows before assignment. This prevents a changed value from being silently attached to the wrong prediction location.

For changed variable q, scalar input is repeated across $\mathcal{J}$, or one replacement q_i^* is supplied for every selected row. The scenario is

$$q_i^{\text{new}} = \begin{cases} q_i^*, & i \in \mathcal{J}, \\ q_i, & i \notin \mathcal{J}. \end{cases} \tag{P-02}$$

Existing factor levels and their ordering are retained. New or missing factor levels fail explicitly rather than becoming a missing design value.

Rebuilding the fitted design and offset

Let T denote the response-free fitted terms object, including its factor levels and contrast definitions. SPmodels independently constructs

$$X = T(\mathcal{D}), \qquad X^* = T(\mathcal{D}^*), \tag{P-03}$$

from the baseline and scenario data. Formula transformations are therefore re-evaluated rather than edited inside a design matrix. Likewise, an in-formula offset expression produces o and o^* from the two model frames. For example, changing exposure E_i in `offset(log(exposure))` changes the offset from $\log E_i$ to $\log E_i^*$ exactly once.

The ordinary link-scale baseline and scenario means have the common fitted parameter vector and common conditional spatial contribution c_i:

$$\eta_i = x_i^{\mathsf{T}} \hat{\beta} + o_i + c_i, \qquad \eta_i^* = x_i^{*\mathsf{T}} \hat{\beta} + o_i^* + c_i. \tag{P-04}$$

The spatial contribution is common because coordinates are immutable in this operation. SPmodels calls the package prediction method for both designs, so Gaussian response predictions use P-04, while binomial and Poisson response predictions apply the already derived uncertainty integration rather than only transforming the link mean. Requested standard errors are recalculated for each design; they are not copied from the baseline.

The reported scenario contrast is descriptive:

$$\Delta_i = \widehat{y}_i^* - \widehat{y}_i. \tag{P-05}$$

It is conditional on the same fitted model and must not be interpreted as a causal effect without a separate identification argument and appropriate model.

Preservation contract

Before prediction, the implementation verifies all of the following:

1. row identifiers and row order are identical;
2. every unmentioned data column is exactly unchanged;
3. coordinate matrices are exactly unchanged;
4. `sf` geometry is exactly unchanged when present; and
5. the fitted object is never modified or refitted.

Coordinate variables cannot be included in `changes`. A location-change exercise is a new prediction design, not a covariate scenario, and should be supplied explicitly as separate prediction data.

Code and validation map

Contract	Implementation	Independent checks
P-01–P-02	`.sp_update_rows()`, `.sp_update_values()`	Numeric, logical and exact named-row mapping; reordered named values; invalid and empty selections
P-03	`sp_prediction_update()` and `.sp_prediction_design()`	Direct manual reconstruction of updated data, design and log-exposure offset; factor-level preservation
P-04–P-05	baseline and scenario calls to `predict()`	Direct equality with two independent prediction calls, including standard errors and differences
Preservation contract	explicit returned audit	Unchanged extra columns, coordinates, row order, `sf` geometry and fitted parameter values

The implementation and tests are in `R/prediction-update.R` and `tests/test-prediction-update.R`.

Grouped random intercepts

Purpose, assumptions and current boundary

Measurements can share both spatial dependence and membership of a clinic, survey, laboratory or other grouping unit. A Gaussian random intercept assigns one common zero-mean deviation to every observation in the same level. SPmodels supports one or more additive grouping variables in Gaussian, binomial-logit and Poisson-log spatial models through `random = ~ group1 + group2`.

The interface derived in this chapter assumes independent Gaussian intercepts across levels and grouping variables. Fixed grouped slopes and correlated intercept-slope effects use the separate explicit interface derived in Chapter 17; neither interface claims arbitrary mixed-model formula syntax or non-spatial grouped fits. Non-Gaussian grouped models support MCML correction. Prediction at a new level is available only through an explicit marginal policy; it is never silently treated as a known level.

Model and indicator matrices

For observations $i = 1, \dots, n$, let x_i contain fixed-effect covariates and let S_i be the Matérn spatial effect. Suppose grouping variable r has L_r observed levels. Its indicator matrix Z_r has entries

$$(Z_r)_{i\ell} = \mathbb{1}\{\text{observation } i \text{ belongs to level } \ell\}. \tag{Q-01}$$

Every row has exactly one unit entry for each grouping variable. The model is

$$Y = X\beta + o + S + \sum_{r=1}^{R} Z_r b_r + \varepsilon, \tag{Q-02}$$

with mutually independent components

$$S \sim N(0, \sigma_s^2 R_\theta), \qquad b_r \sim N(0, \omega_r^2 I_{L_r}), \qquad \varepsilon \sim N(0, \tau^2 I_n). \tag{Q-03}$$

Here ω_r^2 is the variance of the random intercept for grouping variable r. It is not the variance of a group mean and is not a residual variance. Integrating out the random intercepts gives

$$V = \sigma_s^2 R_\theta + \sum_{r=1}^{R} \omega_r^2 Z_r Z_r^\mathsf{T} + \tau^2 I_n. \tag{Q-04}$$

Thus two observations in the same level of grouping variable r receive additional covariance ω_r^2; observations in different levels do not. Contributions from multiple grouping variables add.

Maximum likelihood

For fixed variance parameters, the generalized least-squares estimate is

$$\hat{\beta} = (X^\mathsf{T} V^{-1} X)^{-1} X^\mathsf{T} V^{-1} (y - o). \tag{Q-05}$$

Substitution into the Gaussian log likelihood gives

$$\ell = -\frac{1}{2}\left[n \log(2\pi) + \log|V| + (y - o - X\hat{\beta})^\mathsf{T} V^{-1} (y - o - X\hat{\beta})\right]. \tag{Q-06}$$

SPmodels evaluates (Q-05)–(Q-06) using Cholesky solves, never an explicit matrix inverse. When estimation is requested, every strictly positive covariance parameter, including each ω_r^2, is optimized on its log scale. With fixed covariance controls, one positive random variance must be provided for each grouping variable. A scalar is repeated; a named vector must match the grouping-variable names exactly.

The observed optimizer Hessian is inverted on the log scale when positive definite. The delta method maps the random-variance block back to the ordinary variance scale. An unusable Hessian produces missing standard errors, not invented certainty.

Prediction at known grouping levels

For new prediction rows with fixed-effect matrix X_*, coordinates and known-level indicator matrices Z_{r*}, define the training-to-new covariance

$$C = \sigma_s^2 R_{\theta,*} + \sum_{r=1}^{R} \omega_r^2 Z_r Z_{r*}^{\mathsf{T}}, \tag{Q-07}$$

and the new-to-new latent covariance

$$V_*^{(0)} = \sigma_s^2 R_{\theta,**} + \sum_{r=1}^{R} \omega_r^2 Z_{r*} Z_{r*}^{\mathsf{T}}. \tag{Q-08}$$

The universal-kriging mean remains

$$\hat{y}_* = X_* \hat{\beta} + o_* + C^{\mathsf{T}} V^{-1} (y - o - X\hat{\beta}), \tag{Q-09}$$

and the joint latent prediction covariance is

$$\begin{aligned} \Sigma_* = V_*^{(0)} &- C^{\mathsf{T}} V^{-1} C \\ &+ (X_* - C^{\mathsf{T}} V^{-1} X)(X^{\mathsf{T}} V^{-1} X)^{-1} (X_* - C^{\mathsf{T}} V^{-1} X)^{\mathsf{T}}. \end{aligned} \tag{Q-10}$$

Observation prediction adds $\tau^2 I$. Joint draws, pointwise standard errors and conditional means all use the same covariance. By default, new group levels are rejected because their conditional relationship to fitted levels differs from known-level prediction.

Explicit marginal prediction at new grouping levels

Under `unseen = "marginal"`, a previously unseen level receives a fresh zero-mean random intercept from the fitted distribution. For grouping variable r, split the new-row indicators into Z_{r*}^K, whose columns refer to fitted levels, and U_{r*}, whose columns refer only to distinct unseen levels in the prediction data. Then

$$C = \sigma_s^2 R_{\theta,*} + \sum_{r=1}^{R} \omega_r^2 Z_r (Z_{r*}^K)^{\mathsf{T}}, \tag{Q-10a}$$

because a new random intercept is independent of every training intercept, whereas the new-to-new covariance becomes

$$V_*^{(0)} = \sigma_s^2 R_{\theta,**} + \sum_{r=1}^{R} \omega_r^2 \left\{ Z_{r*}^K (Z_{r*}^K)^{\mathsf{T}} + U_{r*} U_{r*}^{\mathsf{T}} \right\}. \tag{Q-10b}$$

Consequently, a new level does not shift the conditional mean: its marginal mean is zero. Its variance ω_r^2 is retained, however, and two prediction rows carrying the same new level have covariance ω_r^2. Rows carrying distinct new levels have zero covariance from that grouping variable. Substituting (Q-10a)–(Q-10b) into (Q-10) or (Q-15) gives the Gaussian or non-Gaussian predictive covariance respectively:

$$\mathrm{Cov}(b_{r*}^{\mathrm{new}}) = \omega_r^2 U_{r*} U_{r*}^{\mathsf{T}}. \tag{Q-10c}$$

This is a population-distribution prediction for new levels, not an estimate of any particular unseen level's realized intercept. Parameter uncertainty, when requested, rebuilds (Q-10a)–(Q-10c) with the variance from each parameter draw.

Simulation and variograms

Unconditional fitted simulation draws

$$Y^{(b)} = X\hat{\beta} + o + L^{\mathsf{T}} z^{(b)}, \qquad L^{\mathsf{T}} L = V, \quad z^{(b)} \sim N(0, I). \tag{Q-11}$$

For two distinct observations i, j, grouping variable r contributes to the model semivariogram

$$\gamma_{r,ij} = \omega_r^2 \mathbb{1}\{g_{ir} \neq g_{jr}\}. \tag{Q-12}$$

Because this term depends on group membership rather than distance alone, SPmodels averages (Q-12) over the actual pairs in each distance bin and adds it to the Matérn-plus-nugget fitted curve. This is more faithful than pretending a group effect is an isotropic function of distance.

Binomial and Poisson latent models

For a supported non-Gaussian response, grouped effects enter the link-scale latent predictor:

$$\eta = X\beta + o + S + \sum_{r=1}^{R} Z_r b_r. \tag{Q-13}$$

After integrating the independent Gaussian intercepts, the latent covariance is

$$\Sigma = \sigma_s^2 R_\theta + \sum_{r=1}^{R} \omega_r^2 Z_r Z_r^{\mathsf{T}} + \tau^2 I. \tag{Q-14}$$

The Laplace mode, curvature and approximate likelihood in Chapter 5 follow after replacing its covariance by (Q-14). Fixed random variances are supplied in `sp_control()`; when covariance estimation is enabled their logarithms are optimized jointly, and usable Hessian curvature is delta-transformed.

For known new levels, cross and new-new covariance use the grouped terms in (Q-07)–(Q-08). If C is the resulting cross-covariance and H^{-1} the Laplace mode covariance, the approximate link covariance is

$$\Sigma_* = V_*^{(0)} - C^{\mathsf{T}}\Sigma^{-1}C + (\Sigma^{-1}C)^{\mathsf{T}} H^{-1} (\Sigma^{-1}C). \tag{Q-15}$$

Unconditional simulation draws the complete latent vector then applies its response distribution:

$$S_Z^{(b)} \sim N(0, \Sigma), \qquad Y_i^{(b)} \mid S_{Z,i}^{(b)} \sim F_i\{x_i^{\mathsf{T}}\hat{\beta} + o_i + S_{Z,i}^{(b)}\}. \tag{Q-16}$$

MCML uses the same covariance (Q-14), with estimated random variances included on the log scale in its common-random-number objective. Predictive parameter draws rebuild the complete latent covariance for every draw. Laplace fitting, MCML, prediction, simulation, diagnostics, validation and reporting therefore share one grouping contract.

Code and validation map

Equations	Implementation	Independent checks
Q-01–Q-04	`.sp_random_training()`, `.sp_random_covariance()`, `.sp_covariance()`	Two grouping variables; direct indicator matrices and full covariance equality
Q-05–Q-06	`sp_fit()`	Direct GLS coefficients and Gaussian log likelihood
Q-07–Q-10	`predict()` and `sp_predict_draws()`	Direct known-level cross-covariance, mean, point variance and joint covariance
Q-11	`simulate()`	Exact seeded Cholesky reconstruction
Q-12	`sp_variogram()`	Direct pairwise group-difference contribution in every bin
Q-13–Q-14	non-Gaussian `sp_fit()`	Direct latent covariance with fixed variance; positive fitted variance and Hessian schema
Q-15	non-Gaussian prediction methods	Direct point mean/variance and full joint link covariance at known levels
Q-16	`simulate()`	Exact seeded latent-covariance and binomial transform
Q-10a–Q-10c	`.sp_random_new()`, `.sp_random_new_covariance()`, `predict()`, `sp_predict_draws()`	Direct Gaussian and binomial marginal means, point variances and joint covariance for shared unseen levels
stochastic extensions	`sp_mcml()`, `sp_predict_draws()`	Fixed and estimated grouped MCML paths; finite seeded parameter draws with rebuilt covariance
Workflow retention	validation and reporting methods	Fold-wise refitting, simulation recovery, random-variance table rows, safe default failures and explicit marginal unseen-level prediction

The primary checks are in `tests/test-random-effects.R` and `tests/test-nongaussian-random-effects.R`.

Decay-adjusted intervention histories and impact estimation

Purpose, evidence and boundary

Repeated prevalence surveys may occur before and after interventions in the same implementation unit. Treating intervention count as an ordinary linear covariate does not express an immediate reduction followed by decay of that reduction. This chapter derives the history adjustment and constrained impact estimation implemented by `sp_intervention()`, `sp_intervention_multiplier()` and the binomial path in `sp_fit()`.

The model is based on the mathematical specification in Giorgi, Fronterre and Diggle (2026), *A decay-adjusted spatio-temporal model to account for the impact of mass drug administration on neglected tropical disease prevalence*, JRSS A, doi:10.1093/jrsssa/qnag100; the version checked during implementation was arXiv:2512.03760v3. SPmodels' interface, derivation and code are independently written.

The immediate reduction and decay time may be fixed or estimated jointly with regression and spatial covariance parameters by either the labelled Laplace or MCML route. A normalized beta penalty for the immediate reduction is optional. Separable residual temporal correlation is available as derived in Chapter 15.

Data, timing and assumptions

For survey row $i = 1, \dots, n$, define:

- Y_i: positive individuals among m_i examined;
- x_i: fixed-effect covariates;
- t_i: finite numeric survey time;
- g_i: implementation-unit identifier;
- u_{gj}: time of intervention j in unit g;
- $I_{gj} \in \{0, 1\}$: whether that intervention was delivered.

Time differences and the decay parameter must use the same units. An intervention contributes only when $u_{gj} < t_i$. Consequently a survey and intervention recorded at the same time are interpreted as survey first, intervention second. Unit/time pairs in the history must be unique.

Without intervention, baseline prevalence is

$$p_i^* = \text{logit}^{-1}(x_i^\mathsf{T}\beta + o_i + S_i), \qquad S \sim N(0, \Sigma_\psi). \tag{I-01}$$

Here o_i is a known formula offset and Σ_ψ is the existing Matérn-plus-independent latent covariance.

One intervention and the accumulated multiplier

Let $\alpha \in [0, 1]$ be the immediate proportional reduction and $\gamma > 0$ the decay time. At lag $v > 0$, the remaining reduction is

$$f(v) = \alpha \exp(-v/\gamma). \tag{I-02}$$

The corresponding prevalence multiplier is (1-f(v)). For survey row (i), SPmodels multiplies the contributions from all earlier delivered interventions in the same unit:

$$A_i = \prod_{j: u_{g_i j} < t_i} \left[1 - \alpha \exp\{-(t_i - u_{g_i j})/\gamma\}\right]^{I_{g_i j}}. \tag{I-03}$$

An empty product is one. Since every lag in the product is strictly positive, each factor is in ((0,1]), including when $\alpha = 1$; hence $0 < A_i \leq 1$. The observed prevalence and sampling model are

$$p_i = A_i p_i^*, \qquad Y_i \mid S_i \sim \text{Binomial}(m_i, p_i). \tag{I-04}$$

This is not equivalent to adding $\log A_i$ to a logit predictor because $A_i \,\text{logit}^{-1}(\eta_i) \neq \text{logit}^{-1}(\eta_i + \log A_i)$.

Latent score and curvature

Write $q_i = \text{logit}^{-1}(\eta_i)$, $p_i = A_i q_i$, and $q_i' = q_i(1-q_i)$, where $\eta_i = x_i^\mathsf{T}\beta + o_i + S_i$. Apart from constants, observation (i)'s log likelihood is

$$\ell_i = Y_i\{\log A_i + \log q_i\} + (m_i - Y_i)\log(1 - A_i q_i). \tag{I-05}$$

Differentiating once gives

$$\frac{\partial \ell_i}{\partial \eta_i} = Y_i(1-q_i) - (m_i - Y_i)\frac{A_i q_i(1-q_i)}{1 - A_i q_i}. \tag{I-06}$$

For the second term in (I-06), the quotient derivative is

$$\frac{\partial}{\partial \eta_i}\frac{A_i q_i(1-q_i)}{1 - A_i q_i} = \frac{A_i q_i(1-q_i)\{1 - 2q_i + A_i q_i^2\}}{(1 - A_i q_i)^2}. \tag{I-07}$$

Therefore the negative observation curvature is

$$w_i = Y_i q_i(1-q_i) + (m_i - Y_i)\frac{A_i q_i(1-q_i)\{1 - 2q_i + A_i q_i^2\}}{(1 - A_i q_i)^2}. \tag{I-08}$$

When $A_i = 1$, (I-06) becomes $Y_i - m_i q_i$ and (I-08) becomes $m_i q_i(1-q_i)$, recovering ordinary binomial-logit expressions.

Let $P = \Sigma_\psi^{-1}$. The posterior score and negative Hessian for the latent vector are

$$g_i = \frac{\partial \ell_i}{\partial \eta_i} - (PS)_i, \qquad H = P + \text{diag}(w_1, \ldots, w_n). \tag{I-09}$$

SPmodels uses the existing damped Newton iteration when H is positive definite and accepts a step only when the log posterior increases. The direct test differentiates (I-05) numerically and checks both (I-06) and (I-08).

Laplace fitting and MCML

For trial values of α and γ, the latent mode $\widehat{S}$, local covariance H^{-1}, and Laplace log likelihood are calculated exactly as in the non-Gaussian derivation, but using (I-04)–(I-09). Regression and, when requested, covariance parameters are then optimized. Fixed intervention parameters add no fitted degrees of freedom.

For MCML proposal draw S_b, the observation part of the log weight becomes

$$\sum_i \log \text{Binomial}\{Y_i; m_i, A_i \,\text{logit}^{-1}(x_i^\mathsf{T}\beta + o_i + S_{bi})\}. \tag{I-10}$$

The normalized Gaussian proposal, prior density, common-random-number rule and diagnostics remain those in the MCML chapter.

When intervention parameters are estimated, MCML uses

$$q_\alpha = \text{logit}(\alpha), \qquad q_\gamma = \log\gamma. \tag{I-10a}$$

It recalculates every A_i for every trial parameter vector and maximizes

$$\widehat{\ell}_{MC} = \log\left\{ B^{-1} \sum_{b=1}^{B} w_b(\beta, \psi, \alpha, \gamma) \right\} + \log \pi(\alpha), \tag{I-10b}$$

where $\log \pi(\alpha)$ is zero when no beta penalty was declared. Adding the normalized penalty to every log weight is algebraically identical to the last term in (I-10b). Observed MCML curvature is transformed back with

$$J = \operatorname{diag}\{\alpha(1-\alpha), \gamma\}, \qquad V_{\alpha\gamma} = J V_{q_\alpha q_\gamma} J^{\mathsf{T}}. \tag{I-10c}$$

Prediction and simulation

For a prediction row, SPmodels first reconstructs $A_\star$ from its unit, survey time and the retained history. Let the approximate predictive link be $Z_\star \sim N(\mu_\star, v_\star)$. The latent response mean is

$$E(P_\star) = A_\star E\{\text{logit}^{-1}(Z_\star)\}, \tag{I-11}$$

evaluated by the same numerical Gaussian integration used for ordinary binomial prediction. Its latent response variance uses

$$\operatorname{Var}(P_\star) = A_\star^2 \left[E\{\text{logit}^{-1}(Z_\star)^2\} - E\{\text{logit}^{-1}(Z_\star)\}^2 \right]. \tag{I-12}$$

For a future Bernoulli observation, the marginal variance is $E(P_\star)\{1-E(P_\star)\}$. Joint latent response draws transform each Gaussian link draw $z_\star^{(b)}$ as

$$p_\star^{(b)} = A_\star \, \text{logit}^{-1}(z_\star^{(b)}). \tag{I-13}$$

When parameter uncertainty is requested, each Hessian draw contains $(\beta, q_\alpha, q_\gamma)$ and any estimated covariance parameters. For draw b, SPmodels reconstructs

$$\alpha^{(b)} = \text{logit}^{-1}(q_\alpha^{(b)}), \qquad \gamma^{(b)} = \exp(q_\gamma^{(b)}), \qquad A_*^{(b)} = A_*(\alpha^{(b)}, \gamma^{(b)}), \tag{I-13a}$$

then rebuilds the fitted latent state and multiplies that draw's response probability by $A_*^{(b)}$. The returned multiplier matrix makes this transformation auditable.

At fitted survey sites, unconditional simulation draws $S^{(b)} \sim N(0, \Sigma_\psi)$ and then

$$Y_i^{(b)} \sim \text{Binomial}\left[m_i, A_i \, \text{logit}^{-1}\{x_i^{\mathsf{T}}\widehat{\beta} + o_i + S_i^{(b)}\} \right]. \tag{I-14}$$

Cross-validation and simulation recovery retain the complete history while excluding held-out responses from fitting. Thus intervention schedules are treated as known design information, not outcomes that leak across folds.

Algorithm and validation map

1. Validate and normalize unique unit/time intervention rows.
2. Match survey unit and numeric time columns by their declared names.
3. Apply the strict-before rule and calculate (I-03) without rounding.
4. Evaluate the adjusted latent mode through (I-05)–(I-09).

5. Fit regression, covariance and optionally constrained intervention parameters by the declared Laplace or normalized MCML criterion.
6. Recalculate multipliers for new prediction rows; apply (I-11)–(I-14).
7. Retain the normalized history, parameters, timing rule and row multipliers in auditable objects.

Equations	Implementation	Independent checks
I-02–I-04	history constructor and multiplier	Hand-calculated no-event, same-time, one-event, repeated-event and untreated cases
I-05–I-09	`sp_latent_mode()` intervention path	Centred finite-difference score and Hessian; ordinary-binomial limiting case
I-10–I-10c	`sp_mcml()` intervention path	Direct response, prior, proposal and beta-penalty log weights; constraints, multiplier update and delta covariance
I-11–I-13a	`predict()`, `sp_predict_draws()`	Direct integration, exact seeded transform and parameter-draw-specific multiplier reconstruction
I-14	`simulate()`	Exact seeded latent Gaussian and binomial reconstruction
I-15–I-17	constrained Laplace optimizer and reporting	Parameter constraints, normalized penalty reconstruction, degrees of freedom and delta-method covariance
I-18–I-19	effect-curve calculation and plot	Direct curve values, gradient variance, bounds and rendered display
Workflow	validation and reporting methods	Retained intervention specification, fitted/fixed table rows and explicit invalid-input failures

Constrained impact estimation and uncertainty

When estimation is requested, supplied $\alpha \in (0,1)$ and $\gamma > 0$ are starting values. SPmodels optimizes

$$\rho = \log\{\alpha/(1-\alpha)\}, \qquad \xi = \log\gamma, \quad \alpha = \text{logit}^{-1}(\rho), \quad \gamma = e^{\xi}. \tag{I-15}$$

Finite computational bounds keep α away from exact zero and one and scale broad positive decay bounds to the observed history/survey time span. If positive shapes a, b are supplied, the optimized criterion is

$$\ell_{\text{pen}} = \ell_{\text{Laplace}} + \log\left\{\frac{\alpha^{a-1}(1-\alpha)^{b-1}}{B(a,b)}\right\}. \tag{I-16}$$

The beta normalizing constant is retained. The result is labelled *Penalized Laplace*, not an unpenalized marginal likelihood. Let $V_{\rho\xi}$ be the inverse observed-Hessian block. Ordinary-scale uncertainty is

$$V_{\alpha\gamma} = JV_{\rho\xi}J^{\mathsf{T}}, \qquad J = \mathrm{diag}\{\alpha(1-\alpha), \gamma\}. \tag{I-17}$$

An unusable Hessian produces missing standard errors rather than certainty.

Intervention-effect curve and display

For a single intervention and lag $v \geq 0$, the displayed reduction and prevalence multiplier are

$$r(v) = \alpha e^{-v/\gamma}, \qquad m(v) = 1 - r(v). \tag{I-18}$$

For an estimated fit,

$$\nabla r(v) = \left(e^{-v/\gamma}, \quad r(v)\frac{v}{\gamma^2}\right)^{\mathsf{T}}, \qquad \mathrm{Var}\{r(v)\} \approx \nabla r(v)^{\mathsf{T}} V_{\alpha\gamma} \nabla r(v). \tag{I-19}$$

Pointwise normal intervals are clipped to $[0, 1]$; multiplier limits reverse because $m = 1 - r$. The returned table retains unrounded values, and the plot is only a view over that table.

The primary tests are in `tests/test-intervention.R`. They reconstruct the penalized objective, constraints, degrees of freedom, curve uncertainty and display. A fixed or fitted intervention history can coexist with the separable temporal covariance in Chapter 15. The MCML checks reconstruct every final importance log weight and the fitted row multipliers.

Separable space-time geostatistical models

Purpose, data and assumptions

Repeated surveys can be close in space but separated in time. SPmodels represents this residual dependence by multiplying Matérn spatial correlation by exponential temporal correlation. Each point observation needs one two-dimensional projected location and one finite numeric time.

The model assumes stationarity and isotropy in space, stationarity in time, separability, common covariance parameters across the study, and independent observation-specific nugget variation. It does not claim non-separable interaction, directional anisotropy, seasonality or non-stationarity.

From coordinates and times to correlation

For locations $s_i, s_j \in \mathbb{R}^2$ and times t_i, t_j, define

$$d_{ij} = \|s_i - s_j\|, \qquad u_{ij} = |t_i - t_j|. \tag{T-01}$$

Let $R_M(d; \phi, \nu)$ be the Matérn correlation derived in Chapter 1. With temporal scale $\lambda > 0$,

$$R_T(u; \lambda) = \exp(-u/\lambda), \tag{T-02}$$

and the separable correlation is

$$R_{ST}\{(s_i,t_i),(s_j,t_j)\} = R_M(d_{ij};\phi,\nu)R_T(u_{ij};\lambda). \tag{T-03}$$

For finite correlation matrices R_S and R_T,

$$R_{ST} = R_S \circ R_T. \tag{T-04}$$

The Schur product theorem makes this elementwise product positive semidefinite. Its diagonal is one, so it is a valid correlation matrix. At zero time lag it reduces to Matérn correlation; at zero spatial distance it reduces to exponential temporal correlation.

Gaussian model and likelihood

For a Gaussian response,

$$Y = X\beta + o + Zb + S + \epsilon, \tag{T-05}$$

where Zb contains any grouped random intercepts. With spatial variance σ_s^2, nugget variance σ_e^2, and grouped-effect covariance ZGZ^T,

$$V = \sigma_s^2 R_{ST} + ZGZ^\mathsf{T} + \sigma_e^2 I. \tag{T-06}$$

For fixed covariance parameters, generalized least squares gives

$$\hat{\beta} = (X^\mathsf{T}V^{-1}X)^{-1}X^\mathsf{T}V^{-1}(Y-o). \tag{T-07}$$

Writing $e = Y - o - X\hat{\beta}$, the profiled ordinary log-likelihood is

$$\ell = -\tfrac{1}{2}\{n\log(2\pi) + \log|V| + e^\mathsf{T}V^{-1}e\}. \tag{T-08}$$

SPmodels uses Cholesky solves. If temporal scale is estimated, it optimizes $\log\lambda$ jointly with log spatial scale and log variances. The delta method transforms usable observed-Hessian uncertainty back to its ordinary scale:

$$\mathrm{Var}(\hat{\lambda}) \approx \hat{\lambda}^2\,\mathrm{Var}\{\log(\hat{\lambda})\}. \tag{T-09}$$

Indefinite curvature is reported as missing uncertainty, not repaired by weakening a numerical criterion.

Prediction at new locations and times

For new design X_*, times t_*, and locations s_*, the training-to-new covariance has elements

$$(C_*)_{ij} = \sigma_s^2 R_M(\|s_i - s_{*j}\|;\phi,\nu)\exp(-|t_i - t_{*j}|/\lambda), \tag{T-10}$$

plus any known-level grouped-effect covariance. The conditional mean is

$$\widehat{Y}_* = X_*\hat{\beta} + o_* + C_*^\mathsf{T}V^{-1}e. \tag{T-11}$$

Let K_{**} contain all new-to-new space-time covariances and define

$$Q = X_* - C_*^{\mathsf{T}} V^{-1} X. \tag{T-12}$$

The joint latent predictive covariance is

$$\Sigma_* = K_{**} - C_*^{\mathsf{T}} V^{-1} C_* + Q(X^{\mathsf{T}} V^{-1} X)^{-1} Q^{\mathsf{T}}. \tag{T-13}$$

Future-observation prediction adds $\sigma_e^2 I$. Point standard errors use the diagonal of (T-13); joint draws use its full matrix. Optional Gaussian parameter draws include $\log \lambda$ when its Hessian covariance is usable.

Binomial and Poisson models

For binomial-logit and Poisson-log fits, the latent Gaussian prior is

$$\Sigma = \sigma_s^2 R_{ST} + \sigma_e^2 I. \tag{T-14}$$

The Chapter 5 Laplace derivation is unchanged after substituting (T-14). For each trial value of $\log \lambda$, SPmodels rebuilds (T-14), finds the latent mode, and includes both the response-plus-prior value and determinant correction in the Laplace objective. Thus temporal scale is estimated jointly with the regression and spatial covariance parameters rather than from a separate empirical correlation step. Equation (T-09) gives its natural-scale local curvature uncertainty when the full observed Hessian is usable.

MCML keeps the same transformed parameter in its deterministic fixed-draw objective. At every trial value it evaluates the exact Gaussian prior density with the corresponding space-time covariance before dividing by the fixed normalized proposal density. The final latent state, prediction, simulation, validation refits and parameter-recovery summaries all use the resulting $\hat{\lambda}$. This extends the likelihood calculation; it does not remove possible weak identification caused by sparse or confounded survey times.

Simulation, refitting and variograms

Gaussian simulation at fitted sites uses

$$Y^{(b)} = X\hat{\beta} + o + L^{\mathsf{T}} z^{(b)}, \quad z^{(b)} \sim N(0, I), \quad L^{\mathsf{T}} L = V. \tag{T-15}$$

Non-Gaussian simulation transforms a draw from (T-14) through its response model. Validation and simulation-refitting pass the temporal specification to every refit and read held-out times from prediction data.

For Gaussian pair (i, j), the fitted process-plus-nugget semivariance is

$$\gamma_{ij} = \sigma_e^2 + \sigma_s^2\{1 - R_M(d_{ij}; \phi, \nu) \exp(-u_{ij}/\lambda)\}. \tag{T-16}$$

A distance bin can contain different time lags, so SPmodels averages the pair-specific fitted values rather than using only a distance midpoint:

$$\hat{\gamma}_k = |B_k|^{-1} \sum_{(i,j) \in B_k} \gamma_{ij}. \tag{T-17}$$

Grouped random-intercept pair contributions are added as in Chapter 13. Parametric variogram envelopes simulate from the fitted space-time covariance.

Units, interpretation and identifiable limits

Spatial scale uses coordinate units; temporal scale uses the declared `time_units`. Multiplying all times and λ by the same positive constant leaves correlation unchanged. A large temporal scale means slow decay, not a hard duration. If all times coincide, temporal scale is not identifiable, so an estimation request is rejected explicitly. Sparse time coverage or distance-time confounding can likewise produce weak curvature, so estimates require study-design judgment.

Implementation and independent checks

Equations	Implementation	Independent checks
T-01–T-04	`spacetime_cor()`, `sp_temporal()`	Exponential and limiting cases, unit rescaling, symmetry and eigenvalue PSD
T-05–T-09	`sp_fit()`, covariance/state helpers	Direct dense GLS, determinant likelihood, positive scale and Hessian schema
T-10–T-13	`predict()`, `sp_predict_draws()`	Direct cross/new covariance, point variance and joint covariance at new times
T-14	non-Gaussian fitting, prediction and MCML	Direct fitted covariance, positive log-scale estimate, fitted degree of freedom, MCML propagation and finite response/joint prediction
T-15	`simulate()`	Exact seeded Cholesky reconstruction
T-16–T-17	`sp_variogram()`, envelope code	Direct pairwise time-aware bin averages
refitting/reporting	validation and table methods	Specification retained; scale status and units exposed

Primitive checks are in `tests/test-spacetime.R`; fitted-workflow checks are in `tests/test-spacetime-fit.R`. These establish the stated equations for small, well-conditioned fixtures. They do not establish external reference agreement, broad scientific validity or large-data performance.

Transformed local parameter confidence intervals

Purpose and interpretation

`confint()` summarizes local fitted-parameter uncertainty without allowing a Wald interval to cross a parameter's mathematical support. These are large-sample intervals derived from observed-Hessian standard errors, not exact confidence sets, likelihood-profile intervals, boundary corrections or Monte Carlo error intervals.

Let $\hat{\theta}$ have standard error s_θ, and let $z = z_{(1+c)/2}$ for confidence level c.

Unconstrained coefficients

Mean coefficients have real-line support, so their interval is

$$\hat{\theta} \pm z s_\theta. \tag{H-01}$$

Positive parameters

Variances, spatial and temporal scales, random-effect variances and decay are positive. With $g(\theta) = \log \theta$, the delta method gives

$$\mathrm{se}\{g(\hat{\theta})\} \approx s_\theta / \hat{\theta}. \tag{H-02}$$

Transforming a symmetric log-scale interval back gives

$$\left[\exp\{\log \hat{\theta} - z s_\theta / \hat{\theta}\}, \exp\{\log \hat{\theta} + z s_\theta / \hat{\theta}\}\right]. \tag{H-03}$$

Both limits are positive; ordinary-scale asymmetry is intentional.

Intervention impact constrained to zero–one

For $0 < \alpha < 1$, use $g(\alpha) = \mathrm{logit}(\alpha)$. Since

$$g'(\alpha) = \{\alpha(1-\alpha)\}^{-1}, \tag{H-04}$$

the transformed standard error is

$$s_g = s_\alpha / \{\hat{\alpha}(1-\hat{\alpha})\}. \tag{H-05}$$

The ordinary-scale interval is

$$\left[\mathrm{logit}^{-1}\{g(\hat{\alpha}) - z s_g\}, \mathrm{logit}^{-1}\{g(\hat{\alpha}) + z s_g\}\right], \tag{H-06}$$

which lies strictly between zero and one.

Selection, missing curvature and checks

Only estimated parameters appear. Qualified identifiers such as `mean::x`, `covariance::scale`, `intervention::alpha` and `temporal::scale` avoid ambiguous names. Selection is exact.

If an observed Hessian cannot provide a finite standard error, both limits remain missing. SPmodels does not silently substitute curvature. Weak identification, boundary estimates and small samples can make the local normal approximation unreliable.

Equations	Implementation	Independent checks
H-01	`confint.spmodels_fit()`	Direct coefficient interval
H-02–H-03	positive branch	Direct nugget, decay and temporal-scale transforms; positivity

Equations	Implementation	Independent checks
H-04–H-06	impact branch	Direct logit reconstruction and zero–one support
selection/failure policy	qualified rows	Fixed omission, ordering and invalid-selection errors

The checks are in `tests/test-confint.R`.

Grouped random slopes and constrained covariance estimation

Scientific question and current scope

Repeated measurements from the same clinic, school, village or instrument may share more than a common level shift. Their relationship with a numeric covariate can also vary by group. For example, clinics may differ in both baseline prevalence and the change associated with age. SPmodels represents that variation with a Gaussian random intercept and slope whose covariance is declared explicitly with `sp_random_slope()`. The matrix may remain fixed or serve as the starting value for constrained maximum-likelihood estimation.

This implementation supports one or more additive grouped components in all three spatial response families. Each component may contain a slope alone or a correlated intercept and slope. Fixed matrices remain explicit sensitivity assumptions. Estimated matrices use a Cholesky parameterization that is positive definite at every optimizer evaluation; their natural-scale entries, delta-method standard errors and fitted degrees of freedom are reported.

Prerequisites are matrix multiplication, covariance, multivariate normal distributions, generalized least squares, and the prediction derivations in Chapters 2, 5, 6 and 13.

Notation and model

Let there be n observations and R grouped slope components. Component r has grouping value g_{ir}, numeric slope value q_{ir}, and coefficient vector

$$b_{r\ell} = \begin{cases} (b_{0,r\ell}, b_{1,r\ell})^{\mathsf{T}}, & \text{intercept included,} \\ b_{1,r\ell}, & \text{slope only,} \end{cases} \qquad b_{r\ell} \sim N(0, G_r). \tag{RS-01}$$

Different levels ℓ, different components, the spatial process and the nugget are mutually independent. The row vector multiplying the coefficient is

$$w_{ir}^{\mathsf{T}} = \begin{cases} (1, q_{ir}), & \text{intercept included,} \\ q_{ir}, & \text{slope only.} \end{cases} \tag{RS-02}$$

For an intercept and slope,

$$G_r = \begin{pmatrix} \omega_{0r}^2 & \omega_{01,r} \\ \omega_{01,r} & \omega_{1r}^2 \end{pmatrix}. \tag{RS-03}$$

SPmodels requires G_r to be finite, symmetric and positive definite. Positive definiteness implies positive marginal variances and

$$|\omega_{01,r}| < \omega_{0r}\omega_{1r}, \tag{RS-04}$$

so the implied correlation lies strictly between minus one and one.

The Gaussian response model is

$$Y_i = x_i^\mathsf{T}\beta + o_i + S_i + \sum_{r=1}^{R} w_{ir}^\mathsf{T} b_{r,g_{ir}} + \varepsilon_i. \tag{RS-05}$$

Deriving the grouped covariance

The mean of a grouped contribution is zero. For observations i, j, if $g_{ir} \neq g_{jr}$, their coefficient vectors are independent and the covariance is zero. If they share a level, they multiply the same vector. Using $\operatorname{Cov}(Ab, Cb) = A\operatorname{Cov}(b)C^\mathsf{T}$,

$$\begin{aligned} K_{r,ij} &= \operatorname{Cov}(w_{ir}^\mathsf{T} b_{r,g_{ir}}, w_{jr}^\mathsf{T} b_{r,g_{jr}}) \\ &= \mathbb{1}(g_{ir} = g_{jr}) w_{ir}^\mathsf{T} G_r w_{jr}. \end{aligned} \tag{RS-06}$$

For the two-coefficient case, expanding the multiplication gives

$$K_{r,ij} = \mathbb{1}(g_{ir} = g_{jr})\{\omega_{0r}^2 + (q_{ir} + q_{jr})\omega_{01,r} + q_{ir}q_{jr}\omega_{1r}^2\}. \tag{RS-07}$$

This intermediate expression is useful for hand checks. It shows why the group contribution is not constant when slopes are present.

Let K_r be the matrix with entries (RS-06). Independence lets covariance components add, so the integrated Gaussian covariance is

$$V = \sigma_s^2 R_\theta + \sum_r K_r + \sum_a \omega_a^2 Z_a Z_a^\mathsf{T} + \tau^2 I. \tag{RS-08}$$

The final sum permits the independent random intercepts from Chapter 13 to be used simultaneously. Because every G_r is positive definite, each K_r is positive semidefinite: it can be written as a block design matrix times $I \otimes G_r$ times its transpose. Adding it therefore preserves a valid covariance model.

Likelihood and computation

After integration, the exact Gaussian log likelihood and GLS coefficient estimate are precisely equations Q-05 and Q-06 with V from (RS-08). SPmodels forms each K_r directly from the equality matrix $\mathbb{1}(g_{ir} = g_{jr})$ and the product $W_r G_r W_r^\mathsf{T}$, then adds the two element by element. The complete covariance is factorized by Cholesky decomposition for solves and its log determinant.

When `estimate = FALSE`, the supplied G_r entries do not enter the optimizer. Consequently there is no delta-method block for that fixed matrix, and confidence intervals are not fabricated for a supplied assumption.

Positive-definite estimation

For an estimated intercept-slope matrix, write

$$L_r = \begin{pmatrix} \exp(a_r) & 0 \\ c_r & \exp(d_r) \end{pmatrix}, \qquad G_r = L_r L_r^{\mathsf{T}}. \tag{RS-08a}$$

The diagonal exponentials are strictly positive. Because L_r is nonsingular, G_r is positive definite for every finite (a_r, c_r, d_r); the optimizer cannot step to a negative variance or an impossible correlation. Multiplication gives the natural parameters

$$\omega_{0r}^2 = e^{2a_r}, \qquad \omega_{01,r} = e^{a_r} c_r, \qquad \omega_{1r}^2 = c_r^2 + e^{2d_r}. \tag{RS-08b}$$

A slope-only component uses $G_r = e^{2a_r}$. The supplied covariance is converted to its unique Cholesky factor to obtain optimizer starting values. For an exact Gaussian model, these transformed parameters join the other covariance parameters in the profiled likelihood. For a latent non-Gaussian model they join the regression and covariance parameters in the Laplace or fixed-draw MCML objective.

Let $\psi_r = (a_r, c_r, d_r)^{\mathsf{T}}$ and $g_r = (\omega_{0r}^2, \omega_{01,r}, \omega_{1r}^2)^{\mathsf{T}}$. The Jacobian used to return Hessian uncertainty to the readable scale is

$$J_r = \frac{\partial g_r}{\partial \psi_r^{\mathsf{T}}} = \begin{pmatrix} 2e^{2a_r} & 0 & 0 \\ e^{a_r} c_r & e^{a_r} & 0 \\ 0 & 2c_r & 2e^{2d_r} \end{pmatrix}. \tag{RS-08c}$$

If $H_{\psi_r}^{-1}$ is the relevant block of usable inverse observed curvature, the local natural-scale covariance is

$$\operatorname{Var}(\hat{g}_r) \approx J_r H_{\psi_r}^{-1} J_r^{\mathsf{T}}. \tag{RS-08d}$$

Variance intervals use a log transform; the covariance entry, which may be negative, uses a local identity-scale interval. These remain local Wald summaries and can be unreliable near a boundary. For identifiability, an estimated intercept-slope component requires at least three observed groups and at least two distinct slope values within every group. It cannot coexist with a separately estimated random intercept for the same grouping variable, and only one estimated slope component may use that grouping variable. A slope-only component requires at least two groups and at least two non-zero slope observations within each group, so its shared grouped covariance is actually represented in the data.

For binomial-logit and Poisson-log models, the grouped contribution is on the latent link scale. Replacing the latent covariance in Chapter 5 by (RS-08) produces the implemented Laplace mode, curvature and objective. MCML uses the same covariance in the Gaussian prior density; no hidden secondary grouping calculation is used.

Prediction at known and unseen levels

For training row i and prediction row j, define

$$C_{r,ij} = \mathbb{1}(g_{ir} = g_{jr}^*) w_{ir}^{\mathsf{T}} G_r w_{jr}^*. \tag{RS-09}$$

The new-to-new component is

$$K^*_{r,jk} = \mathbb{1}(g^*_{jr} = g^*_{kr})(w^*_{jr})^\mathsf{T} G_r w^*_{kr}. \tag{RS-10}$$

These terms are added to the spatial cross and new covariance before applying the Gaussian universal-kriging equations Q-09–Q-10 or the non-Gaussian approximation Q-15.

An unseen level is rejected by default. With `unseen = "marginal"`, it is a fresh zero-mean coefficient vector independent of all training groups. Therefore its column in (RS-09) is zero. Equation (RS-10) still applies among prediction rows: two rows with the same new label share one new coefficient vector, while distinct new labels are independent. The slope values of both rows remain in the covariance, so shared-new-level dependence is generally not constant.

The same policy is explicit in cross-validation. If a group-constrained fold withholds an entire fitted grouping level, the default prediction error is a useful guard against silently treating it as known. Passing `unseen = "marginal"` to `sp_cross_validate()` instead evaluates the declared new-group prediction problem and carries that policy into integrated non-Gaussian log scores.

Simulation and variograms

Unconditional simulation uses one draw from the complete covariance (RS-08). This is distributionally identical to drawing each group coefficient vector separately and adding $w_{ir}^\mathsf{T} b_{r,g_{ir}}$, because sums of independent Gaussian vectors remain Gaussian and have additive covariance.

For any pair, the grouped semivariance contribution is

$$\gamma_{r,ij} = \tfrac{1}{2}(K_{r,ii} + K_{r,jj}) - K_{r,ij}. \tag{RS-11}$$

It depends on group membership and both covariate values, not only distance. SPmodels therefore evaluates (RS-11) for the actual observation pairs and averages it within each variogram distance bin.

Small worked calculation

Suppose two observations in clinic A have slope values 1 and 2 and

$$G = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}.$$

Their row vectors are $w_1 = (1, 1)$ and $w_2 = (1, 2)$. Equation (RS-06) gives

$$K_{12} = (1, 1)G(1, 2)^\mathsf{T} = (1, 1)(6, 19)^\mathsf{T} = 25.$$

The individual variances are $K_{11} = 15$ and $K_{22} = 44$, so their grouped semivariance is $(15 + 44)/2 - 25 = 4.5$. A row in clinic B has covariance zero with both, even if its slope value is identical.

Assumptions, checks and limitations

- Group labels and numeric slope values must be observed without missingness.
- Covariates should be centered when the intercept-slope covariance is to have a useful interpretation; shifting the zero of a slope changes the meaning of the random intercept and their covariance.

- A fixed G_r is a sensitivity assumption. Analysts should repeat fits over scientifically defensible matrices when it is uncertain. An estimated G_r needs enough independent groups and within-group covariate variation; passing the minimum checks does not by itself prove strong identification.
- The current method has one numeric slope per component, no random nonlinear basis, and no non-spatial fit.
- Agreement tests independently construct (RS-06)–(RS-10), the GLS estimate, likelihood, joint prediction covariance, Laplace covariance and MCML state.
- For fitted-parameter predictive draws, every sampled Cholesky parameter set rebuilds both the training and prediction-side copies of (G_r); using the fitted-centre matrix for the new-site block would mix two parameter states.

Practice

1. Expand (RS-06) to obtain (RS-07).
2. Show that a slope-only model has covariance $\mathbb{1}(g_i = g_j)q_i q_j \omega_1^2$.
3. Explain why a new level changes predictive variance but not conditional mean under marginal prediction.
4. Recalculate the worked semivariance after centering the slope values at 1.5, remembering that a scientifically equivalent covariance matrix would also need reparameterisation.

Code and validation map

Equations	Implementation	Independent check
RS-01–RS-04	`sp_random_slope()`	dimensions, symmetry and Cholesky failures
RS-05–RS-08	`.sp_random_slope_training()`, `.sp_random_slope_covariance()`, `sp_fit()`	direct full covariance, GLS and log likelihood
RS-08a–RS-08d	Cholesky parameter helpers, Gaussian/Laplace/MCML fitting, `confint()`	reconstruction, positive definiteness, analytic-versus-numerical Jacobian, natural-scale uncertainty and fitted degrees of freedom
RS-09–RS-10	`predict()`, `sp_predict_draws()`	direct cross and joint covariance; shared unseen levels
parameter-state propagation	`.sp_prediction_random_slope()`	deterministic perturbed Cholesky state appears identically in the prediction-side covariance
RS-08 in latent models	Laplace fitting and `sp_mcml()`	exact latent-prior covariance equality
RS-11	`sp_variogram()`	direct pairwise semivariance contribution
reporting contract	`summary()`, `sp_model_table()`	fixed versus estimated status, natural-scale standard errors and likelihood degrees of freedom

The main executable checks are in `tests/test-random-slopes.R`.

Bounded-memory point prediction

Purpose and exactness

A prediction grid may contain many more rows than the fitted dataset. Ordinary prediction forms a training-by-new cross-covariance matrix. For n training rows and m prediction rows this has nm entries, even though pointwise means and marginal standard errors do not require covariance between distinct prediction rows.

`sp_predict_chunked()` partitions the new rows and calls the same fitted-model prediction equations on each partition. It changes memory scheduling only. It does not approximate distances, covariance, latent conditioning or nonlinear response integration.

Partition and equivalence

Let the ordered new-row indices be partitioned into disjoint consecutive sets

$$I_1 \cup \cdots \cup I_J = \{1, \ldots, m\}, \qquad I_j \cap I_k = \emptyset \quad (j \neq k), \tag{CH-01}$$

with at most c rows per set. For any set I_j, the ordinary Gaussian conditional mean is

$$\hat{y}_{I_j} = X_{I_j}\hat{\beta} + o_{I_j} + C_{I_j}^{\mathsf{T}} V^{-1}(y - o - X\hat{\beta}), \tag{CH-02}$$

and its pointwise variances are the diagonal entries of equation Q-10. No column of C_{I_j} enters the mean or marginal variance of a row outside I_j. Concatenating results in original order therefore gives

$$(\hat{y}_{I_1}^{\mathsf{T}}, \ldots, \hat{y}_{I_J}^{\mathsf{T}})^{\mathsf{T}} = \hat{y}_* \tag{CH-03}$$

and the same equality holds for the diagonal predictive variances.

For binomial and Poisson models, equations CH-01–CH-03 apply first to the approximate latent mean and variance in Chapter 5. Logistic-normal quadrature or lognormal moments are rowwise transformations of those quantities, so the response means and marginal standard errors are also unchanged.

Memory and limitations

The unchunked cross covariance needs $O(nm)$ numeric storage. A chunk needs at most $O(nc)$, plus output vectors of length m. Computation still evaluates all nm distances and therefore has the same leading arithmetic cost. Chunking does not make dense model fitting scalable in n.

The method must not be used as a substitute for joint prediction. Covariance between rows in different chunks is not constructed. Regional targets, simultaneous excursions and dependent grid simulations must continue to use `sp_predict_draws()` or a future explicitly validated scalable joint method.

Known and marginal unseen grouping levels retain ordinary pointwise behavior. Rows sharing a new level across chunks would have joint covariance, but this does not alter either row's marginal mean or variance; that cross-row covariance is simply outside this function's declared output.

Algorithm and checks

1. Validate the fitted object, non-empty prediction data and positive chunk size.
2. Form consecutive index sets by $\lceil i/c \rceil$.
3. Call `predict()` with identical target, scale and unseen-level policy for each set.
4. Concatenate means and standard errors in original row order and restore row identifiers.
5. Record the requested chunk size and number of chunks.

`tests/test-scalable-prediction.R` compares chunk sizes that do not divide the row count against one-shot Gaussian and binomial predictions, including observation uncertainty, response integration, fixed correlated grouped slopes and a shared unseen level. Equality is checked numerically, along with ordering, metadata and invalid-input failures.

Bounded-memory nearest-predecessor predictive draws

Purpose and approximation boundary

Chapter 6 derives the fitted joint Gaussian predictive distribution

$$\eta_* \mid y \dot{\sim} N(m, S), \tag{NP-01}$$

where the dot is needed for non-Gaussian fits because their latent prediction already uses a Laplace approximation. Ordinary joint drawing constructs the complete $M \times M$ matrix S. That is exact for the declared predictive distribution but requires quadratic grid memory.

`sp_predict_draws_neighbor()` instead defines a sparse sequential factorization. It is exact when every predecessor is retained and otherwise a clearly labelled, order-dependent approximation. It reduces prediction-grid working memory; it does not make the dense training fit scalable and does not include fitted-parameter uncertainty.

From the Gaussian chain rule to local conditioning

Choose a permutation $\pi = (\pi_1, \dots, \pi_M)$ of prediction rows. The probability chain rule is the identity

$$p(\eta_{\pi_1}, \dots, \eta_{\pi_M}) = \prod_{j=1}^{M} p(\eta_{\pi_j} \mid \eta_{\pi_1}, \dots, \eta_{\pi_{j-1}}). \tag{NP-02}$$

For row $i = \pi_j$, let $P_i = \{\pi_1, \dots, \pi_{j-1}\}$. If all of P_i is retained, the multivariate-normal conditioning identity gives

$$\eta_i \mid \eta_{P_i} \sim N(a_i, v_i), \tag{NP-03}$$

$$a_i = m_i + S_{iP_i} S_{P_i P_i}^{-1} (\eta_{P_i} - m_{P_i}), \tag{NP-04}$$

$$v_i = S_{ii} - S_{iP_i} S_{P_i P_i}^{-1} S_{P_i i}. \tag{NP-05}$$

Multiplying (NP-03) over all rows recovers exactly the density in (NP-01). Sequential simulation from these conditionals is therefore another factorization of the same Gaussian distribution, analogous to Cholesky simulation.

Nearest-predecessor approximation

Fix a positive neighbor budget r. For every row i, SPmodels selects

$$N_i = \text{the at most } r \text{ members of } P_i \text{ having smallest Euclidean coordinate distance to } i. \quad \text{(NP-06)}$$

Ties retain predecessor order. It then replaces (NP-03) by

$$\tilde{p}(\eta_*) = \prod_{j=1}^{M} p(\eta_{\pi_j} \mid \eta_{N_{\pi_j}}), \quad \text{(NP-07)}$$

using (NP-04)–(NP-05) with P_i replaced by N_i. For independent standard-normal values $z_i^{(b)}$, draw b is generated recursively as

$$\eta_i^{(b)} = \tilde{a}_i^{(b)} + \sqrt{\tilde{v}_i}\, z_i^{(b)}. \quad \text{(NP-08)}$$

The predictive submatrix for $N_i \cup \{i\}$ is calculated directly from the same equations as the ordinary joint method. This retains spatial, temporal and supported grouped covariance contributions inside each local conditional. Cholesky solves are used in (NP-04)–(NP-05), not explicit matrix inversion. Coincident latent prediction rows can make the predecessor block positive semidefinite. If Cholesky fails, a checked symmetric eigendecomposition uses the Moore–Penrose solution on eigenvalues above a scale-dependent tolerance and verifies that the cross-covariance lies in the predecessor column space. It does not add artificial jitter. A materially negative eigenvalue, incompatible cross-covariance or conditional variance is an error; only round-off-sized negative conditional variance is set to zero.

If $r \geq M - 1$, then $N_i = P_i$ for every row and (NP-07) equals (NP-02). This exact boundary is both a mathematical check and an executable test. For smaller r, omitted long-range conditional links change the joint distribution. Marginal means are retained, but neither exact covariances nor exact regional uncertainty should be claimed.

Ordering, grouped effects and interpretation

The default permutation is input-row order. A caller may supply an explicit permutation or request a deterministic maximin order. Maximin starts with input row one. If A_{j-1} is the set already selected, the next row is

$$\pi_j = \operatorname*{arg\,max}_{i \notin A_{j-1}} \left\{ \min_{k \in A_{j-1}} \|c_i - c_k\|_2 \right\}, \quad \text{(NP-06a)}$$

with input order breaking ties. The running minimum distance requires only $O(M)$ storage, although selecting the complete order uses $O(M^2)$ distance work. Returned draw columns always use original row order. Because the neighbor sets in (NP-06) depend on the permutation, approximate results can change with ordering. Sensitivity to defensible orders and neighbor counts should be assessed for scientific analyses.

Neighbor selection uses spatial coordinate distance. A temporal or grouped effect still enters the selected local covariance, but it does not change which predecessors are selected. In particular, a far-away row sharing a grouped level may be omitted. Increasing r, choosing an informed order or using the ordinary exact joint method is necessary when such long-range structure drives the target.

Gaussian response/link draws use (NP-08) directly. Binomial-logit and Poisson-log link draws use their Chapter 6 local Gaussian predictive approximation, followed by the same inverse-link and optional response-sampling steps as ordinary joint draws.

Optional fitted-parameter uncertainty

Chapter 6 equation (J-15) defines the fitted-Hessian parameter draw $\psi^{(b)}$. When `parameter_uncertainty = TRUE`, SPmodels rebuilds the fitted covariance or Laplace state for each such draw. Denote its predictive moments by $m^{(b)}$ and $S^{(b)}$. Equations (NP-04)–(NP-08) are then evaluated using those draw-specific moments:

$$\eta_i^{(b)} = m_i^{(b)} + S_{iN_i}^{(b)}\{S_{N_iN_i}^{(b)}\}^{-1}(\eta_{N_i}^{(b)} - m_{N_i}^{(b)}) + \sqrt{\tilde{v}_i^{(b)}}\, z_i^{(b)}. \tag{NP-09}$$

Estimated intervention impact and decay rebuild the row multipliers for every response-scale draw. The parameter draws and multiplier draws are retained. The reported conditional variance is the average of the draw-specific local conditional variances; no complete predictive covariance is manufactured. When every predecessor is retained, this uses the same parameter and prediction random-number order as ordinary `sp_predict_draws()`, providing a draw-for-draw exact boundary. With a small neighbor budget it combines the local joint approximation with the separate local-Hessian parameter approximation.

Streaming grouped targets

When only grouped weighted targets are required, storing all BM point draws is unnecessary. `sp_predict_target_neighbor()` normalizes the raw target weights exactly as in Chapter 6 and updates

$$T_g^{(b)} \leftarrow T_g^{(b)} + w_{ig} D_i^{(b)}. \tag{NP-10}$$

as soon as point i is simulated. A point may contribute to several targets, as occurs with fractional polygon-cell allocations.

Let $\ell(i)$ be the last processing step at which point i appears in a later neighbor set, or its own step if it is never reused. Its B-draw link vector is deleted immediately after step $\ell(i)$. If W is the largest number of simultaneously active point vectors and G the number of targets, draw storage is of order

$$O\{B(W+G)\}, \tag{NP-11}$$

rather than automatically $O\{B(M+G)\}$. The neighbor graph itself uses $O(Mr)$ indices. The result records W, the graph, order, conditional variances and sparse normalized allocation rows. It does not claim that W is always small: an early site used by many later conditionals remains active, and retaining all predecessors can give $W = M$.

Streaming currently conditions on fitted parameters. For observation-scale binomial or Poisson targets, response sampling occurs when each point is visited, so a fixed seed need not give the same individual response draws as a post-hoc workflow that consumes all Gaussian random numbers first. The declared distribution is unchanged; Gaussian and latent-response cases have direct same-seed aggregation checks.

Analytic implied covariance diagnostic

For a manageable validation subset, the approximation can be examined without Monte Carlo noise. Define the local coefficient row

$$A_i = S_{iN_i} S_{N_i N_i}^{-1}. \tag{NP-12}$$

After approximate covariance rows for all predecessors P_i have been constructed, the recursion implied by (NP-07) is

$$\widetilde{S}_{iP_i} = A_i \widetilde{S}_{N_i P_i}, \tag{NP-13}$$

$$\widetilde{S}_{ii} = \tilde{v}_i + A_i \widetilde{S}_{N_i N_i} A_i^{\mathsf{T}}. \tag{NP-14}$$

`sp_neighbor_covariance()` applies (NP-12)–(NP-14) and can compare the result with the ordinary exact fitted predictive covariance. It reports relative Frobenius error, maximum absolute covariance error and maximum marginal- variance error. This diagnostic intentionally allocates M^2 entries; it is for selecting/checking a budget on manageable subsets, not the production bounded-memory route. With all predecessors, the recursion recovers the exact fitted covariance.

Storage and computation

At a single step, the largest newly constructed cross-covariance has order $n(r+1)$, and the local prediction covariance has order $(r+1)^2$, where n is the number of training rows. The complete M^2 prediction covariance is never returned or allocated. The unavoidable returned draw matrix contains BM values for B draws.

This is a memory result, not a speed guarantee. With parameter uncertainty, the local conditionals are rebuilt for every one of the B parameter draws. Repeated dense training solves can be expensive, and the original fitted model still required dense training covariance factorization. Sparse or low-rank training likelihoods remain a separate development and validation problem.

Algorithm and validation map

1. Validate the spatial fit, draw count, neighbor budget and row permutation.
2. Build the prediction design once and generate all call-local standard normals.
3. Resolve input, maximin (NP-06a) or user-supplied order; visit rows in that order and find (NP-06).
4. Construct only the predictive distribution for $N_i \cup \{i\}$.
5. Calculate (NP-04)–(NP-05) by Cholesky solves and draw with (NP-08), or rebuild the fitted state and use (NP-09) for every parameter draw.
6. Restore original column order, apply any response transformation, and return neighbor indices, conditional variances and order instead of a full covariance matrix.
7. For direct targets, apply (NP-10), release point vectors after their last graph use and report (NP-11)'s observed active width.
8. For an explicit covariance diagnostic, compose (NP-12)–(NP-14) and, optionally, compare it with the exact fitted covariance.

Equations	Implementation	Independent checks
NP-01–NP-05	existing joint predictive equations plus sequential loop	Same-seed equality to ordinary Gaussian and non-Gaussian link draws when all predecessors are retained
NP-06–NP-08	`sp_predict_draws_neighbor()`	Neighbor-count bound, deterministic maximin/user ordering, reordered moments, exact full-predecessor boundary and reproducibility
grouped covariance	local design subsetting	Same-seed exact boundary with a shared unseen marginal group level
NP-09	parameter-draw loop	Same-seed Gaussian and fitted-intervention equality to ordinary all-predecessor joint draws; retained parameter/multiplier draws
NP-10–NP-11	`sp_predict_target_neighbor()`	Same-seed Gaussian and binomial-latent post-hoc target equality, repeated polygon allocation, sparse normalized weights, peak active width and caller-seed preservation
NP-12–NP-14	`sp_neighbor_covariance()`	Exact all-predecessor Gaussian/non-Gaussian equality, PSD duplicate boundary, finite small-neighbor errors and symmetric positive-semidefinite result
failure and audit contracts	public validation and returned metadata	Invalid budget/order, local seed preservation, no full covariance and printed approximation label
singular predecessor boundary	`.sp_psd_solve()`	Coincident latent rows give identical draws and zero second conditional variance without jitter

The executable checks are in `tests/test-neighbor-prediction.R`.

Gaussian nearest-predecessor likelihood

Purpose and evidence boundary

Dense Gaussian fitting factorizes an n-by-n covariance matrix. Its storage grows as $O(n^2)$, and a general Cholesky factorization takes $O(n^3)$ arithmetic. This chapter derives the explicit approximation used by `sp_fit_neighbor()`. It replaces the dense joint density by ordered local conditional densities. It does not change `sp_fit()`, and it is not claimed to be exact unless every predecessor is retained.

The initial implementation supports a Gaussian identity-link point model with one stationary isotropic Matérn field and independent nugget noise. Grouped, random-slope, temporal and non-Gaussian likelihoods require separate derivations and validation and are rejected rather than approximated silently.

The ordered conditional approximation follows the statistical construction of Vecchia (1988). The implementation and equations below were derived directly for SPmodels; no external software implementation was used. Guinness (2018) explains why permutation is part of the approximation and studies ordering and grouping strategies. SPmodels presently implements the stated input and deterministic maximin orders, not the paper's broader grouping methods.

Model and assumptions

For observations $i = 1, \dots, n$, let

$$y_i = x_i^{\mathsf{T}}\beta + o_i + w_i + e_i, \qquad e_i \overset{ind}{\sim} N(0, \tau^2), \tag{NL-01}$$

where o_i is a known offset and

$$\mathrm{Cov}(w_i, w_j) = \sigma^2 \rho_\nu(\|s_i - s_j\|; \phi). \tag{NL-02}$$

The coordinates are projected, both axes use the recorded distance unit, $\sigma^2 > 0$, $\phi > 0$, $\tau^2 > 0$, and the smoothness $\nu > 0$ is fixed. The mean design has full column rank. Write $y_o = y - o$ and let $V_{ij} = \mathrm{Cov}(y_i, y_j)$. Thus

$$V_{ij} = \sigma^2 \rho_\nu(d_{ij}; \phi) + \tau^2 \, \mathbb{1}(i = j). \tag{NL-03}$$

Ordered conditional factorization

Choose a permutation $\pi = (\pi_1, \dots, \pi_n)$. The exact density always factorizes as

$$p(y) = p(y_{\pi_1}) \prod_{r=2}^{n} p(y_{\pi_r} \mid y_{\pi_1}, \dots, y_{\pi_{r-1}}). \tag{NL-04}$$

For row $i = \pi_r$, let $P_i = \{\pi_1, \dots, \pi_{r-1}\}$. Retain the at most m members of P_i nearest to s_i; call that set N_i. The approximation is

$$\tilde{p}(y) = \prod_{r=1}^{n} p(y_{\pi_r} \mid y_{N_{\pi_r}}). \tag{NL-05}$$

For input ordering, $\pi_r = r$. Deterministic maximin ordering starts with row one and repeatedly selects the unselected location farthest from the selected set. A user-supplied permutation is also permitted. Order and graph are retained because they are part of the approximation.

One local Gaussian conditional

For a current observation i, define

$$a_i = V_{iN_i} V_{N_i N_i}^{-1}, \qquad d_i = V_{ii} - a_i V_{N_i i}. \tag{NL-06}$$

Standard multivariate-normal conditioning gives

$$y_{o,i} \mid y_{o,N_i} \sim N\left(x_i^{\mathsf{T}}\beta + a_i(y_{o,N_i} - X_{N_i}\beta),\ d_i\right). \tag{NL-07}$$

No inverse is formed in code. The system in (NL-06) is solved by Cholesky factorization, with the checked positive-semidefinite solve used for an exact singular boundary. A non-positive conditional variance fails the candidate parameter state.

Define transformed data

$$\tilde{y}_i = y_{o,i} - a_i y_{o,N_i}, \qquad \tilde{x}_i^{\mathsf{T}} = x_i^{\mathsf{T}} - a_i X_{N_i}. \tag{NL-08}$$

Equation (NL-07) is now an independent weighted regression statement under the approximate density:

$$\tilde{y}_i \sim N(\tilde{x}_i^{\mathsf{T}}\beta, d_i). \tag{NL-09}$$

Profiled mean and approximate log likelihood

Let $D = \mathrm{diag}(d_1, \ldots, d_n)$, stack the transformed rows in $\widetilde{X}$, and stack the responses in $\tilde{y}$. For fixed covariance parameters $\theta = (\sigma^2, \phi, \tau^2)$, the generalized least-squares estimate is

$$\hat{\beta}(\theta) = (\widetilde{X}^{\mathsf{T}} D^{-1} \widetilde{X})^{-1} \widetilde{X}^{\mathsf{T}} D^{-1} \tilde{y}. \tag{NL-10}$$

Substitution gives the profiled approximate log likelihood

$$\tilde{\ell}(\theta) = -\frac{1}{2} \sum_{i=1}^{n} \left\{ \log(2\pi) + \log d_i + \frac{(\tilde{y}_i - \tilde{x}_i^{\mathsf{T}}\hat{\beta})^2}{d_i} \right\}. \tag{NL-11}$$

The positive covariance parameters are optimized on their log scales with bounded L-BFGS-B. The conditional coefficient covariance reported at the final state is

$$\widehat{\mathrm{Var}}(\hat{\beta} \mid \hat{\theta}) = (\widetilde{X}^{\mathsf{T}} D^{-1} \widetilde{X})^{-1}. \tag{NL-12}$$

When the optimizer Hessian in log covariance parameters is positive definite, its inverse is transformed to the natural scale by the delta method. This is a local curvature approximation; it does not include finite-neighbor model error.

Exact all-predecessor boundary

If $m \geq n-1$, then $N_i = P_i$ for every row. Equation (NL-05) is exactly (NL-04), whatever valid order is used. Consequently (NL-10)–(NL-12) equal the dense Gaussian ML log likelihood, profiled regression estimate and conditional coefficient covariance, apart from floating-point roundoff. This is an executable boundary check, not an empirical expectation.

For $m < n-1$, the output is an approximation. Increasing m need not make every finite-sample parameter estimate move monotonically, so selection must use a predeclared stability study rather than choosing the most convenient result.

Local prediction

For a new location 0, select its at most m nearest training observations N_0. With

$$a_0 = V_{0N_0} V_{N_0N_0}^{-1}, \tag{NL-13}$$

the plug-in conditional mean is

$$\hat{y}_0 = x_0^{\mathsf{T}}\hat{\beta} + o_0 + a_0(y_{o,N_0} - X_{N_0}\hat{\beta}). \tag{NL-14}$$

Let $g_0^{\mathsf{T}} = x_0^{\mathsf{T}} - a_0 X_{N_0}$. The approximate latent prediction variance is

$$\sigma^2 - V_{0N_0} V_{N_0N_0}^{-1} V_{N_00} + g_0^{\mathsf{T}}\widehat{\mathrm{Var}}(\hat{\beta})g_0. \tag{NL-15}$$

Add τ^2 for a future observation. Each point is conditioned locally; the method does not return or imply a joint prediction covariance. Use the separate nearest-predecessor draw and covariance-audit functions when joint approximate prediction is required.

At a training row, the same calculation uses that row's recorded predecessor set. Its mean-estimation gradient is the corresponding transformed design row from (NL-08); this contribution is included in the reported sequential fitted- value standard error. Consequently the observation and latent variances differ by exactly τ^2 at a common row.

Storage, computation and limitations

The retained graph uses $O(nm)$ indices. Each likelihood contribution factorizes at most an m-by-m covariance block, so covariance workspace is $O(m^2)$ rather than $O(n^2)$. The direct implementation takes up to $O(nm^3)$ factorization work per objective evaluation; neighbor discovery is currently an exact distance scan with $O(n^2)$ time but bounded working memory. This is a memory-scalable statistical foundation, not yet a compiled or indexed speed implementation.

The approximation does not remove the need for recovery studies, neighbor- budget sensitivity checks, cross-platform timings and comparison against a separately frozen reference environment. Non-Gaussian, grouped and temporal extensions remain unsupported until their own derivations and tests exist.

Algorithm and validation map

1. Build the model frame, projected coordinates and full-rank mean design.
2. Resolve and retain input, maximin or supplied order and at most m nearest predecessors per row.

3. For each covariance candidate, compute (NL-06), transform with (NL-08), and profile the mean by (NL-10).
4. Optimize (NL-11) on log covariance scales or evaluate supplied fixed values.
5. Retain the graph, local coefficients, conditional variances, optimizer and unrounded estimates.
6. Predict pointwise with (NL-13)–(NL-15), explicitly labelling the result as nearest-predecessor approximate.

Equations	Implementation	Independent checks
NL-01–NL-03	existing Matérn and nugget covariance	fixed-parameter simulated fixture
NL-04–NL-09	local graph and Gaussian-state builders	graph budget/order, positive local variances, caller RNG unchanged
NL-10–NL-12	profiled local GLS and optimizer	all-predecessor dense likelihood, coefficient and covariance equality; finite estimated state
NL-13–NL-15	neighbor-fit prediction method	finite latent/observation predictions and observation variance ordering
failure contract	`sp_fit_neighbor()` validation	invalid budget/order, coincident-only coordinates and unsupported grouped covariance fail explicitly

The executable checks are in `tests/test-neighbor-fit.R`.

References

Vecchia, A. V. (1988). Estimation and model identification for continuous spatial processes. *Journal of the Royal Statistical Society: Series B*, 50(2), 297–312. https://doi.org/10.1111/j.2517-6161.1988.tb01729.x

Guinness, J. (2018). Permutation and grouping methods for sharpening Gaussian process approximations. *Technometrics*, 60(4), 415–429. https://doi.org/10.1080/00401706.2018.1437476